

Worksheet 1.1: Solutions	
Atomic theory	

No.	Answer
1	a 98 b 154
2	${}^{63}_{28}\text{Ni}$
3	a 15 protons, 16 neutrons and 18 electrons b 92 protons, 143 neutrons and 90 electrons
4	${}^{12}_6\text{C}$
5	a They have the same number of protons and electrons. b They react in the same way as they have the same electron configuration.
6	If the number of protons in an atom changes, then the identity of the atom also changes. When an isotope undergoes alpha decay, it loses two protons, that is, it becomes an element with an atomic number two less than the original atomic number. For beta decay, one of the neutrons decomposes to form an electron and a proton, so the newly formed isotope has one more proton than the original isotope.
7	The reaction that occurs is ${}^{241}_{94}\text{Pu} \rightarrow {}^{241}_{95}\text{Am} + \text{e}^{-}$ One of the neutrons in the plutonium-241 nucleus decomposes to form a proton and an electron. The number of protons in the nucleus therefore increases by 1 and the atom becomes an americium atom. The total number of nucleons (protons and neutrons) remains the same, that is, the mass number is unchanged. The electron produced is ejected energetically by the nucleus.
8	${}^{229}_{90}\text{Th} \rightarrow {}^4_2\text{He} + {}^{225}_{88}\text{Ra}$

Worksheet 1.2: Solutions Electron configurations	
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No.	Answer
1	First shell
2	First shell
3	8 electrons
4	a 2, 8, 6 b 2, 5 c 2, 8, 8, 2
5	a 2, 8, 8 b 2, 8 c 2, 8, 8 d 2, 8, 8
6	Neutral atoms of group 14 elements have four electrons in their outermost shell.
7	The electrons in an atom are attracted to the positively charged nucleus. Energy needs to be used to break the attractive force between the electrons and the nucleus.
8	2, 8, 2
9	The first electron is removed from a neutral atom. But the second electron is removed from a positively charged ion (it has a charge of +1) and so this electron will be more strongly attracted to this species than the first electron was attracted to the species it was being removed from.
10	The third electron is being removed from the second shell but the first two electrons were being removed from the third shell. The second shell is closer to the nucleus than the third shell so the third electron will be more strongly attracted to the nucleus than the first two electrons from the third shell.
11	The eleventh electron to be removed from the magnesium atom would come from the first shell. This is much closer, on average, to the nucleus than the second shell. This eleventh electron will therefore be more strongly attracted to the nucleus than the tenth electron, which is in the second shell. As a result, much more energy will be required to remove the eleventh electron compared to the tenth electron.
12	Looking at the ionisation energies there is a large jump in the amount of energy required to remove the fourth electron compared to the energy required to remove the third electron. As a result it appears that the outer three electrons in these atoms are in a shell further from the nucleus than the fourth electron. That is, the atoms have three electrons in their outermost shell, so are from group 13. Of the three given elements, aluminium is the group 13 element with three electrons in its outermost shell (2, 8, 3).

Worksheet 1.3: Solutions Famous scientists	
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No.	Answer
1	Humphry Davy temporarily blinded himself in an explosion, so Faraday became his apprentice. Faraday performed experiments with magnetism. He liquefied chlorine, discovered benzene and helped refine electrolysis.
2	Cavendish was one of the richest men in England, but he never bought anything. He rarely spoke to anyone and never changed his suit. He discovered the composition of water and air, and he worked out the weight of the Earth!
3	Einstein studied the Brownian motion of particles in a liquid. He estimated the number of particles in one mole.
4	Rutherford was an unemployed teacher. He came second in a scholarship competition, but the winner turned the prize down—a trip to England. Transmutation is the conversion of one element to another. Rutherford was the first to do this, with the conversion of nitrogen to oxygen.
5	Italy. He hypothesised that many gases were diatomic, such as H ₂ .
6	Berzelius improved the consistency of the naming of chemical compounds. He introduced the chemical symbols still used today. This led to the writing of chemical equations.
7	Colour-blindness. John Dalton was the first to document this condition after his mother complained of his inability to buy clothes of the correct colour!
8	Marie Curie. Working with her husband Pierre, she isolated the radioactive elements radium and polonium.
9	Geber is the most famous of the alchemists. He documented the properties of many substances and many chemical processes, such as distillation.
10	Francis Aston invented the mass spectrometer in 1919. It gave exact masses of elements, and helped identify isotopes of elements. Exact masses of compounds could be determined, and the composition of compounds established.
11	Rosenberg was executed for treason for passing information to the Russians about the first US nuclear weapons.
12	Bohr suggested that the electrons were ‘jumping’ between shells (energy levels) in the atoms of the element. When these electrons returned to their original shell, light of a specific colour or wavelength was emitted. This emission of light of a specific wavelength provided information about the arrangement of the electrons in the atom.

Worksheet 1.4: Solutions Developing a periodic table	
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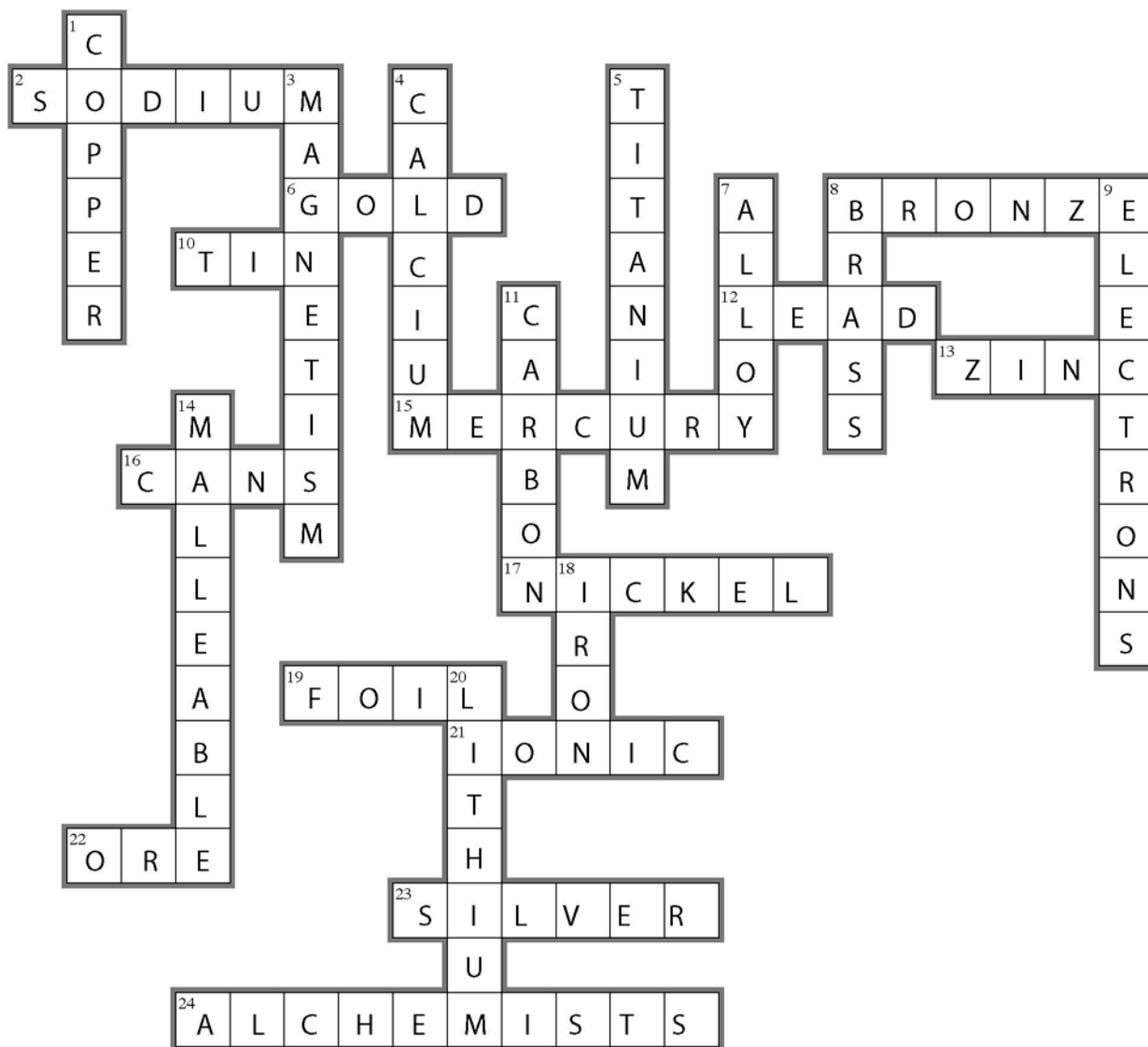
No.	Answer
1	They are examples of the earliest known elements.
2	Elements like Li, Na and K have common chemical properties.
3	No. The modern periodic table uses the atomic number to order the elements.
4	After a while, the elements do not match the groups that they should. The further you go, the greater the inconsistency.
5	He left gaps if no known element seemed to fit.
6	Silicon and group 18 elements.
7	They have the same number of electrons in their outer shells.
8	Argon seemed to sit all on its own—the rest of the group was missing.
9	Elements in the same vertical group have the same outer-shell configuration.
10	There are several alternate forms. Web Destinations on the <i>Companion Website</i> for Chapter 1 lists possible sites to view these tables.

Worksheet 2.1: Solutions

Metals – history, uses and properties

NAME:

CLASS:



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Worksheet 2.1: Solutions

Metals – history, uses and properties

ACROSS

2. SODIUM—The metal ion present in table salt.
6. GOLD—Which metal completes the quote ‘all that glitters is not ...?’
8. BRONZE—The age dating from 3000 BC is named after this metal.
10. TIN—Copper and this metal combine to form bronze.
12. LEAD—A soft, dense metal formerly used in plumbing.
13. ZINC—A reactive metal often used in torch batteries.
15. MERCURY—A liquid metal associated with mad hatters.
16. CANS—Tin-plated iron is used to make these useful containers.
17. NICKEL—Combines with copper to produce Australian ‘silver’ coins.
19. FOIL—This form of aluminium is used in cooking.
21. IONIC—This type of bond exists in the compound formed when a metal reacts with a non-metal.
22. ORE—An economically useful mineral.
23. SILVER—One of the precious metals.
24. ALCHEMISTS—They sought to change lead to gold.

DOWN

1. COPPER— One of the metals that is not silver in colour.
3. MAGNETISM—Iron and some other metals exhibit this property.
4. CALCIUM—An important metal for healthy bones and teeth.
5. TITANIUM—The ninth most abundant element in the Earth’s crust.
7. ALLOY—A mixture of metals.
8. BRASS—An alloy of copper and zinc.
9. ELECTRONS—The conductivity of metals is due to the movement of these particles.
11. CARBON—This element is added to iron to produce steel.
14. MALLEABLE—A property shared by all metals.
18. IRON—The metal used to transport oxygen around your body.
20. LITHIUM—The metal with the smallest atomic number.

Worksheet 2.2: Solutions Modifying metals	
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No.	Answer
1	Ordinary steel corrodes too easily in damp conditions.
2	Tin must be non-toxic and unreactive.
3	Alloys in jewellery give the desirable appearance of the precious metal, but with added strength.
4	Solder must be soft and have a low melting point.
5	Brass is sonorous, relatively light and unreactive. It also has an attractive, shiny appearance.
6	The orderly lattice of positive ions in aluminium are disrupted by the presence of the copper ions. As a result, the layers of positive ions cannot move as easily over one another, making it more difficult to bend the alloyed metal.
7	Titanium is rarer and more expensive than aluminium.
8	Lighter aircraft have reduced fuel consumption, can carry a bigger load and can stay airborne more easily!
9	Wings need to be flexible.
10	During quenching, a metal is heated, then cooled very quickly. Metals are thought to be made up of many crystalline regions. In these regions the ions are neatly lined up, and the layers can easily move over one another. The larger these crystalline areas, the softer the metal. When the hot metal is cooled quickly in the quenching process, only small crystalline regions form, making the metal hard.

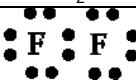
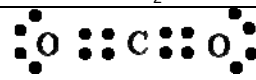
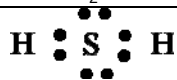
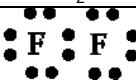
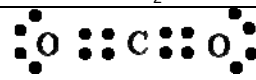
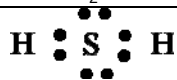
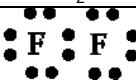
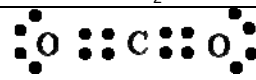
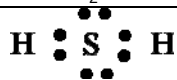
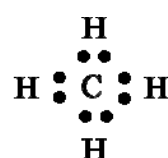
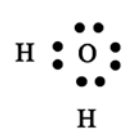
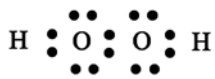
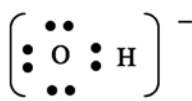
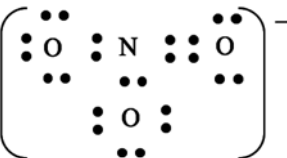
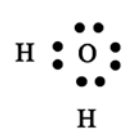
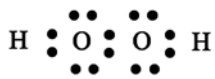
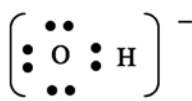
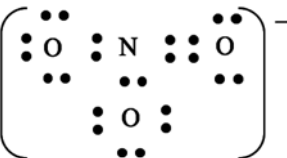
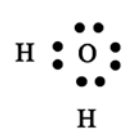
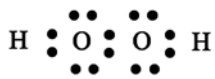
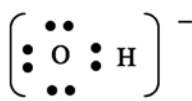
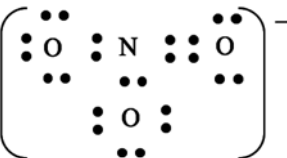
Worksheet 3.1: Solutions Ionic solids	
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No.	Answer
1	b Tin and chlorine c Sodium and fluorine
2	Calcium – lose 2 electrons to form Ca^{2+} Potassium – lose 1 electron to form K^+
3	Chlorine – gain 1 electron to form Cl^- Nitrogen – gain 3 electrons to form N^{3-}
4	a CuF_2 b Al_2O_3
5	A^{3+} bonds with B^- to form AB_3
6	As the charges must be balanced, three A^{2+} ions are required to balance two B^{3-} ions. A is in Group 2, B is in Group 15.
7	a Two Na atoms (2,8,1) each lose one electron to one O atom (2, 6). The ions formed are Na^+ (2, 8) and O^{2-} (2, 8). (see page 67 in the text book for similar electron dot diagrams) b Each Mg atom (2, 8, 2) will lose two electrons to a sulfur atom (2, 8, 6). The ions formed are Mg^{2+} (2, 8) and S^{2-} (2, 8, 8).
8	CaO has the higher melting point as the ionic bond is stronger. This is possibly because the ions in CaO (Ca^{2+} and O^{2-}) have larger charges than the ions in KF (K^+ and F^-). Ions with larger charges will attract one another more strongly than ions with smaller charges.
9	In order for a substance to conduct electricity, it must have charged particles (ions or electrons) that are free to move. The ions that make up solid sodium chloride are trapped in the solid lattice and are not free to move; so it cannot conduct. When dissolved in water (or melted), these ions break away from the rigid lattice and are mobile. As a result, electricity can be conducted.
10	When sufficient force is applied to the NaCl ionic lattice, it is possible to disrupt the alignment of the positive and negative ions in such a way that the same charges line up with each other. The repulsive electrostatic forces cause the crystal to shatter. However, when the lattice of positive ions in sodium is disrupted, the delocalised electrons are still present between the positive ions, maintaining the metallic bonding (attraction between the delocalised electrons and the positive ions) that hold the lattice together. The sodium therefore does not shatter, but it is malleable.

Worksheet 3.2: Solutions Electrovalencies	
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No.	Answer
1	a SO_4^{2-} d NO_2^- b Ca^{2+} e F^- c NO_3^- f Pb^{2+}
2	a chromium(II) ion d cyanide ion b ammonium ion e manganese(II) ion c tin(IV) ion f dihydrogenphosphate ion
3	a carbonate b sulfate c nitrate d phosphate e permanganate All of the names finish in '-ate'.
4	Examples include: fluoride (F^-), chloride (Cl^-), bromide (Br^-), iodide (I^-), hydride (H^-), cyanide (CN^-), oxide (O^{2-}), peroxide (O_2^{2-}), sulfide (S^{2-}), nitride (N^{3-}) Almost all are anions of individual non-metal elements. Most, except oxide and peroxide, do not contain any oxygen atoms.
5	Group 13. X forms a 3+ ion.
6	a KOH d $\text{Cu}_3(\text{PO}_4)_2$ b $\text{Li}_2\text{Cr}_2\text{O}_7$ e $\text{Mg}(\text{HSO}_4)_2$ c SrI_2 f $\text{Al}_2(\text{C}_2\text{O}_4)_3$
7	+2, +3, +4
8	a barium oxide d sodium hydrogencarbonate b copper(II) nitrate e iron(II) dihydrogenphosphate c tin(II) hydroxide f strontium nitride
9	a Ra^{2+} b Te^{6+} c U^{3+} d Au^{3+} e La^{3+}
10	a SeO_3^{2-} selenite ion b CNO^- cyanate ion c BrO_3^- bromate ion

Worksheet 4.1: Solutions Electron dot diagrams of molecules	
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No.	Answer																								
1	<table><tr><th>Compound</th><th>Ionic</th><th>Covalent</th><th>Metallic</th></tr><tr><td>Copper</td><td></td><td></td><td>•</td></tr><tr><td>Oxygen (O₂)</td><td></td><td>•</td><td></td></tr><tr><td>Ammonia (NH₃)</td><td></td><td>•</td><td></td></tr><tr><td>Calcium sulfide</td><td>•</td><td></td><td></td></tr><tr><td>Potassium carbonate</td><td>•</td><td>•</td><td></td></tr></table>	Compound	Ionic	Covalent	Metallic	Copper			•	Oxygen (O ₂)		•		Ammonia (NH ₃)		•		Calcium sulfide	•			Potassium carbonate	•	•	
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2	Only glucose (b) and nitrogen monoxide (d) are molecules. Helium is monatomic, brass is a metal and aluminium chloride is ionic.																								
3	<table><tr><th>Molecule</th><th>F₂</th><th>CO₂</th><th>H₂S</th></tr><tr><td>Electron dot diagram</td><td></td><td></td><td></td></tr><tr><td>Bonding pairs</td><td>1</td><td>4 (2 double)</td><td>2</td></tr><tr><td>Non-bonding pairs</td><td>6</td><td>4</td><td>2</td></tr></table>	Molecule	F ₂	CO ₂	H ₂ S	Electron dot diagram				Bonding pairs	1	4 (2 double)	2	Non-bonding pairs	6	4	2								
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4	Carbon has an electron configuration of 2, 4. It will react with other atoms to gain a share of another four electrons to give it eight electrons in its outermost shell. Each hydrogen atom has one electron. Carbon will therefore combine with four hydrogen atoms to gain a share of eight electrons and to form a molecule with the formula of CH₄:  If each carbon atom only combined with three hydrogen atoms, there would only be 7 electrons in the outermost shell of the carbon atom. If the carbon combined with five hydrogen atoms there would be 9 electrons in the outermost shell. The most stable number of electrons in carbon's outer shell is 8 electrons.																								
5	<table><tr><th>H₂O</th><th>H₂O₂</th><th>OH⁻</th><th>NO₃⁻</th></tr><tr><td></td><td></td><td></td><td></td></tr></table>	H ₂ O	H ₂ O ₂	OH ⁻	NO ₃ ⁻																				
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Worksheet 4.1: Solutions Electron dot diagrams of molecules	
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6	a phosphorus trichloride b trichloromethane c sulfur dioxide d ethyne
7	a O ₂ , as a double bond is longer than a triple bond. b N ₂ , as a triple bond is stronger than a double bond.
8	a $\left[\text{Ca} \right]^{2+} \left(\begin{array}{ccccc} \cdot\cdot & & \cdot\cdot & & \cdot\cdot \\ \cdot\cdot & \text{O} & \cdot\cdot & \text{C} & \cdot\cdot & \text{O} & \cdot\cdot \\ \cdot\cdot & & \cdot\cdot & & \cdot\cdot & & \cdot\cdot \\ & & \cdot\cdot & & \cdot\cdot & & \\ & & \cdot\cdot & \text{O} & \cdot\cdot & & \\ & & \cdot\cdot & & \cdot\cdot & & \end{array} \right)^{2-}$ b Between the positive calcium ions and the negative carbonate ions there is ionic bonding. Between the carbon atoms and the oxygen atoms in the carbonate ions there is covalent bonding.

Worksheet 4.2: Solutions Polythene	
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1	Because of the side chains projecting from the main chain, the long molecules are not able to move very close together. As a result, there are fewer molecules in a particular space, so this form of polythene has less mass in a particular volume. This means it will have a lower density.
2	a Polythene can be thought of as a covalent molecular substance. As a result, there are only weak forces between its molecules. Only a small amount of heat will be required to disrupt these forces, allowing the solid to melt. b In high density polythene, the molecules are able to fit closer together. Consequently, there are stronger intermolecular bonds between the molecules in high density polythene compared to the intermolecular forces in low density polythene. The stronger the intermolecular bonds, the higher the melting point of the solid.
3	If a substance is to bend (be flexible), its molecules need to be able to slip over one another reasonably easily. Because the forces between the chains in high density polythene are stronger than those in low density polythene, it is more difficult for the closer chains to move over one another, leading to a less flexible material.
4	For electrons to conduct electricity they must be able to move through the material. The electrons in polythene are either localised in covalent bonds or in the inner shells of the atoms. There are therefore no ‘free’ electrons able to move under the influence of an electric field.
5	See Figure 4.8b, page 84 in the textbook.
6	a b
7	

Worksheet 5.1: Solutions Covalent network lattices	
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No.	Answer
1	a See Figure 5.7, page 111 in text book. b Between the carbon atoms there are covalent bonds.
2	Silicon (element) and silicon carbide (compound) or silicon dioxide (compound) have similar structures to diamond. They are all extremely hard, have high melting points and are not soluble in water. None of these substances conduct electricity in either the solid or liquid states.
3	See Figure 5.9, page 112 in textbook. Graphite is a layer lattice structure of carbon atoms bonded in an interlocking hexagon arrangement, which is very different to the tetrahedral network lattice of diamond. Each carbon atom only has three covalent bonds. The fourth electron is delocalised between the layers.
4	Graphite is able to conduct electricity because each carbon atom is bonded to only three others, leaving one valence electron free on every carbon atom. These delocalised electrons can move between the layers of hexagonally arranged carbon atoms and so conduct electricity. Diamond has no free electrons, as all four valence electrons are involved in covalent bonds.
5	Diamond is extremely hard because of the very strong covalent bonds, between the carbon atoms, that occur in all three dimensions in its network lattice structure.
6	Diamond is used on drill bits and saws to cut through rock and metal, and can be used to cut glass.
7	Carbon dioxide exists as discrete molecules with only very weak intermolecular forces attracting adjacent molecules to one another. Silicon dioxide exhibits a similar tetrahedral network lattice structure to that of diamond. The strong covalent bonds in all three dimensions ensure that silicon dioxide is very hard and has a high melting point.
8	Elemental silicon exhibits a similar tetrahedral network lattice structure to that of diamond. The strong covalent bonds in all three dimensions ensure that silicon is hard and brittle and has a high melting point. Elemental phosphorus exists in a number of allotropic forms, but mainly as discrete molecules with only weak intermolecular forces between them.
9	Graphite is able to act as a lubricant due to its layer lattice structure. The layers are held together by only weak (dispersion) forces and so they can readily slide past one another, making this substance an excellent lubricant. Its very high melting point means that it is able to withstand the high temperatures generated inside an engine. This high melting point occurs because covalent bonds between the carbon atoms in the layers must be broken when graphite is melted. These bonds are very strong.
10	a Both have large numbers of carbon atoms covalently bonded together in a lattice. b Both have repeating hexagons consisting of six covalently bonded carbon atoms. Nanotubes are similar to a sheet of graphite rolled into a cylinder.

Worksheet 5.2: Solutions

A bonding summary

1	<table><tr><td>Substance</td><td>Ionic</td><td>Covalent</td><td>Metallic</td></tr><tr><td>Aluminium oxide</td><td>•</td><td></td><td></td></tr><tr><td>Hydrogen sulfide</td><td></td><td>•</td><td></td></tr><tr><td>Zinc hydroxide</td><td>•</td><td>•</td><td></td></tr><tr><td>Brass</td><td></td><td></td><td>•</td></tr><tr><td>Graphite</td><td></td><td>•</td><td></td></tr><tr><td>Tungsten</td><td></td><td></td><td>•</td></tr></table>	Substance	Ionic	Covalent	Metallic	Aluminium oxide	•			Hydrogen sulfide		•		Zinc hydroxide	•	•		Brass			•	Graphite		•		Tungsten			•																	
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2	a Metallic b Covalent c Ionic																																													
3	Test the conductivity of the solid. If it conducts, it is a metal (as it is white, it cannot be graphite. Also metals are normally not white unless covered with material from corrosion). If it does not conduct, continue testing: Try to dissolve the solid in water. If it dissolves, test the conductivity of the aqueous solution. If it does not conduct, it is molecular. If it does, it most likely ionic but there is a small possibility that it could be molecular (where the molecules ionise in water). Try to melt the substance. If it melts, test the conductivity. If it conducts, it is ionic.																																													
4	<table><tr><td></td><td>Ionic</td><td>Metallic</td><td>Molecular</td><td>Covalent network</td></tr><tr><td>a</td><td>Positive and negative ions</td><td>Positive ions and electrons</td><td>Molecules</td><td>Atoms</td></tr><tr><td>b</td><td>See figure 3.2, page 54 in textbook</td><td>See figure 2.12, page 37, textbook</td><td>See figure 4.1, p 89 (show both inter- and intramolecular bonds)</td><td>See figure 5.7, page 111, textbook</td></tr><tr><td>c</td><td>Hard, but brittle</td><td>Generally hard, but malleable</td><td>Generally soft</td><td>Very hard</td></tr><tr><td>d</td><td>High</td><td>Generally high</td><td>Low</td><td>Very high</td></tr><tr><td>e</td><td></td><td></td><td></td><td></td></tr><tr><td>s</td><td>No</td><td>Yes</td><td>No</td><td>No</td></tr><tr><td>l</td><td>Yes</td><td>Yes</td><td>No</td><td>No</td></tr><tr><td>aq</td><td>Yes (if soluble)</td><td>Not applicable</td><td>Often no, but sometimes yes (those that ionise)</td><td>Not applicable</td></tr></table>		Ionic	Metallic	Molecular	Covalent network	a	Positive and negative ions	Positive ions and electrons	Molecules	Atoms	b	See figure 3.2, page 54 in textbook	See figure 2.12, page 37, textbook	See figure 4.1, p 89 (show both inter- and intramolecular bonds)	See figure 5.7, page 111, textbook	c	Hard, but brittle	Generally hard, but malleable	Generally soft	Very hard	d	High	Generally high	Low	Very high	e					s	No	Yes	No	No	l	Yes	Yes	No	No	aq	Yes (if soluble)	Not applicable	Often no, but sometimes yes (those that ionise)	Not applicable
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5	a See figure 3.3, page 55 in textbook. b See figure 2.11, page 37 in textbook. c See figure 3.5, page 56 in text book.																																													

Worksheet 6.1: Solutions

Calculation of relative masses

No.	Answer			
1	Complete the table below:			
	Term	Definition	Symbol	Unit
	Relative atomic mass	The weighted mean of the atomic masses of the naturally occurring isotopes on the scale in which an atom of carbon-12 is taken to have a mass of 12 exactly.	A_r or RAM	No unit
	Mole	The amount of substance containing the same number of particles as there are atoms in 12 g of carbon-12.	n	mol
	Molar mass	The mass of one mole of a substance.	M (or M_r)	g mol^{-1}
	Avogadro's constant	The number of elementary particles (6.022×10^{23}) in one mole.	N_A	mol^{-1}
2	The relative atomic mass scale is based on the isotope carbon-12 having a mass of 12 exactly. Carbon occurs naturally as three different isotopes (^{12}C , ^{13}C and ^{14}C). The relative atomic mass of the element is the weighted mean of the masses of these three isotopes.			
3	Because of all the atoms present in a sample of sulfur, most have an atomic mass of 31.97, and only a few have a mass greater than this. The relative atomic mass of sulfur as the element will therefore be approximately 32 (or a little larger than 32).			
4	a 27.028 b 97.994 c 60.062			
5	a 407.68 b 100.09 c 132.144			
6	Relative molecular mass of $\text{NO}_x = 14.01 + 16.00x = 46 \quad \therefore x = 2 \quad \therefore \text{NO}_2$			
7	Relative formula mass of $\text{SnCl}_2 = 189.6$ Relative formula mass of $\text{SnCl}_4 = 260.5$			
8	a 19 times heavier b 19 times heavier c 19 times heavier d 19 g			
9	a 65.38 g mol^{-1} b $136.28 \text{ g mol}^{-1}$			

Worksheet 6.2: Solutions Mole calculations	
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No.	Answer
1	a 239.2 g b 134.45 g c 98.076 g
2	a $n(\text{BaSO}_4) = \frac{m}{M} = \frac{250}{233.36} = 1.07 \text{ mol}$ b $n(\text{Fe}_2\text{O}_3) = \frac{m}{M} = \frac{500000}{159.7} = 3.13 \times 10^3 \text{ mol}$ c $n(\text{C}_6\text{H}_6) = \frac{m}{M} = \frac{3.388 \times 10^{-6}}{78.108} = 4.338 \times 10^{-8} \text{ mol}$ d $n(\text{Al}_2\text{O}_3) = \frac{m}{M} = \frac{30.0 \times 10^6}{101.96} = 2.94 \times 10^5 \text{ mol}$
3	a $N(\text{electrons}) = n \times N_A = 1 \times 6.022 \times 10^{23} = 6.02 \times 10^{23}$ b $N(\text{atoms}) = n \times N_A = 12.2 \times 6.022 \times 10^{23} = 7.35 \times 10^{24}$ c $N(\text{atoms}) = n \times N_A = \frac{120}{39.95} \times 6.022 \times 10^{23} = 1.81 \times 10^{24}$ d $N(\text{molecules}) = n \times N_A = \frac{7.20 \times 10^{-8}}{48.00} \times 6.022 \times 10^{23} = 9.03 \times 10^{14}$
4	a $m(\text{C}_8\text{H}_{18}) = n \times M = 40.0 \times 114.224 = 4570 \text{ g}$ b $m(\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5) = n \times M = 2.80 \times 10^{-5} \times 294.304 = 8.24 \times 10^{-3} \text{ g}$ c $m(\text{Ca}_3(\text{AsO}_4)_2) = n \times M = 3.079 \times 398.08 = 1226 \text{ g}$
5	a $n(\text{H}) = 2.6 \times 2 = 5.2 \text{ mol}$ b $n(\text{H}) = 8.1 \times 10^{-3} \times 14 = 0.11 \text{ mol}$ c $n(\text{BeH}_2) = \frac{m}{M} = \frac{35.0}{11.028} = 3.173 \text{ mol} \therefore n(\text{H}) = 3.173 \times 2 = 6.35 \text{ mol}$ d $n(\text{Al}_2(\text{HPO}_4)_3) = \frac{m}{M} = \frac{977}{341.894} = 2.858 \text{ mol} \therefore n(\text{H}) = 2.858 \times 3 = 8.57 \text{ mol}$
6	$M(\text{XO}_2) = \frac{m}{n} = \frac{55.37}{0.693} = 79.90 \text{ g mol}^{-1}$ $\therefore M(\text{X} + (2 \times 16)) = 79.90 \text{ g mol}^{-1}$ $\therefore M(\text{X}) = 47.9 \text{ g mol}^{-1}$, and so element X is titanium.
7	a $M(\text{C}_5\text{H}_5\text{NO}_2) = 111.1 \text{ g mol}^{-1}$ b $m(\text{C}_5\text{H}_5\text{NO}_2) = n \times M = 8.22 \times 10^{-4} \times 111.1 = 0.0913 \text{ g}$

Worksheet 7.1: Solutions	
The kinetic molecular theory	

No.	Answer
1	a The pressure increases as a result of the increased frequency of collisions with the walls of the container. b Average kinetic energy stays constant because temperature is constant. c Average speed is related to average kinetic energy, so that stays constant too.
2	Increasing the temperature increases the average kinetic energy of the molecules, so they collide more frequently with the walls, and with greater energy. Hence, pressure increases.
3	The oxygen gas will diffuse into the empty flask until the pressure in both flasks is equal.
4	Decreasing the temperature of the gas decreases the average kinetic energy of the molecules. They will thus collide less frequently with the walls of the syringe and with less energy. The plunger moves in until the pressure inside the syringe is equal to the external pressure.
5	a O₂ has the greatest molar mass, so flask B contains the greater mass of gas. b If the flasks are at the same temperature, the molecules will have the same average kinetic energy. c Flask B has the greater density (greater mass in the same volume).
6	See figure 7.2, page 151, textbook. The peak has moved to the right and the spread of velocities is greater.
7	The helium balloon would be more deflated as the smaller He atoms would be able to escape through the holes more easily than the larger N ₂ molecules. With a smaller number of particles inside the balloon, there will be less pressure exerted by the gas in the balloon, so the balloon containing the helium will deflate more rapidly.
8	At very low temperatures, when the molecules are moving slowly, and at very high pressures, when the molecules are forced close together, intermolecular forces become significant.
9	As the balloon rises the atmospheric pressure outside the balloon decreases. Because the size of the balloon is determined by the 'inside' gas pressure being equal to the outside (atmospheric) pressure, the balloon's volume will increase to cause a decrease in the 'inside' pressure. (When the balloon's volume increases, there is a greater surface area so the gas particles will collide less often with this surface and the pressure will be decreased.)

Worksheet 8.1: Solutions Water – a glossary	
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No.	Some suggested diagrams/equations
1	You could show the particles in two liquids, one of pure water and the other with sodium ions and chloride ions in water. With fewer water particles at the surface of (or for bubbles within) the NaCl solution, the vapour pressure of this solution can be shown to be less than that of water. As a result, the NaCl solution needs to be heated to a higher temperature before its vapour pressure equals the atmospheric pressure and the bubbles, of H ₂ O vapour, recognised as boiling can form in the liquid.
2	See figure 5.19, page 121 in the textbook. This shows the equipment set-up for distillation. If separating two liquids by distillation, the liquid with the lower boiling point will boil first and will be the first fraction to be collected.
3	Diagrams showing the ions in the mixture of silver chloride in water still organised in their solid lattice and the separated ions in the solution of sodium chloride could be drawn. Perhaps some indication that the ions in the second mixture would move if electricity (shown as electrodes + power source) was passed through the solution could also be represented in the diagram.
4	Equation: $\text{HCl(g)} + \text{aq} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$. A diagram showing HCl molecules entering water and being ionised into the two ions. You might also include a diagram showing the movement of the electrons from the HCl covalent bond to the chlorine atom to form the chloride ion.
5	Possibly the particles in the two liquids could be drawn with more particles having evaporated (that is, in the gas phase above the liquid) for the ethanol. A reference to the weaker bonds between the ethanol molecules could also be included.
6	You might show the particles in some perspiration evaporating from the skin. Arrows could be used to show the flow of heat energy from the skin to the evaporating water as it changes state and cools down (the more energetic water particles evaporate, leaving the ‘colder’ particles behind).
7	You might use sketches of beakers of boiling water, complete with thermometers, in both a high-altitude situation (a mountain top) and a sea-level situation (on a beach). The boiling points are different. Perhaps show fewer particles in the boiling bubbles for the liquid on the mountain (and reference to it exerting small vapour pressure), and fewer particles in the atmosphere.
8	A diagram showing two separate layers of liquids could be shown, as seen in Figure 5.18, page 121 in the textbook. They form the two separate layers because they are not soluble ‘in one another’.
9	Possibly draw a diagram showing the particles in both gas and liquid phases during boiling, then labelled arrows showing the transforming of the heat energy into potential energy as the particles break away from the attractive forces in the liquid to move further apart in the gas phase. Reference could be made to the kinetic energy of the particles not being changed by this heat energy.

Worksheet 8.1: Solutions Water – a glossary	
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10	Diagrams showing the ions in copper carbonate remaining joined together when added to water but the ions in copper sulfate moving apart in water (and a label of 'blue' for the copper ions) could be drawn.
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Worksheet 8.2: Solutions Solubility curves	
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No.	Answer
1	40 g/100 g of water
2	Potassium chlorate
3	Potassium iodide
4	Potassium iodide
5	Solubility at 30°C (to produce a saturated solution) = 47 g/100 g of water, so the solution is unsaturated.
6	Solubility at 40°C = 198 g/100 g of water. Mass undissolved = 300 – 198 = 102 g.
7	At 40°C, the solubility is 198 g in 100 g of water. We have 102 g to dissolve in x g of water. $\therefore x = \frac{102}{198} \times 100 = 51.5 \text{ g} \therefore 51.5 \text{ mL of water must be added}$
8	Solubility at 80°C = 172 g/100 g water, or 86 g dissolved in 50 g of water Solubility at 20°C = 33 g/100 g water, or 16.5 g dissolved in 50 g of water Mass that crystallises out = 86 – 16.5 = 69.5 g
9	Solubility at 50°C = 215 g of KI per 100 g of water We have 100 g of KI to dissolve in x g of water. $\therefore x = \frac{100}{215} \times 100 = 46.5 \text{ g}$
10	Solubility = $\frac{9.0}{60} \times 100 = 15 \text{ g/100 g of water.}$ This corresponds to a minimum temperature of approximately 35°C to dissolve all the solid (that is, to form a saturated solution).

Worksheet 8.3: Solutions Concentration calculations	
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No.	Answer
1	$V(\text{ethanol}) = \frac{14}{100} \times 700 \text{ mL} = 98 \text{ mL}$
2	$n(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = \frac{m}{M} = \frac{40.0}{342.296} = 0.1169 \text{ mol}$ $\therefore c(\text{C}_{12}\text{H}_{22}\text{O}_{11}) = \frac{n}{V} = \frac{0.1169}{0.300} = 0.390 \text{ mol L}^{-1}$
3	$n(\text{of NaCl in original solution}) = \frac{m}{M} = \frac{70.2}{58.544} = 1.199 \text{ mol} = n(\text{of NaCl in dilute solution})$ $V(\text{dilute NaCl solution}) = \frac{n}{c} = \frac{1.199}{0.670} = 1.790 \text{ L}$ $\therefore (1.790 - 1.00) = 0.790 \text{ L of water must be added.}$
4	$m(\text{NH}_3 \text{ in dilute solution}) = \frac{1.50}{100} \times 250 = 3.75 \text{ g} = m(\text{NH}_3 \text{ in concentrated solution})$ $V(\text{concentrated solution}) = \frac{100}{5.00} \times 3.75 = 75.0 \text{ mL}$
5	$n(\text{Na}_2\text{CO}_3 \text{ in dilute solution}) = c \times V = 1.08 \times 0.0300 = 0.03240$ $n(\text{Na}_2\text{CO}_3 \text{ in concentrated solution}) = 0.03240$ $m(\text{Na}_2\text{CO}_3) = n \times M = 0.03240 \times 105.99 = 3.43 \text{ g}$
6	$m(\text{glucose in 1}^{\text{st}} \text{ solution}) = \frac{2.0}{100} \times 50 = 1.00 \text{ g}$ $m(\text{glucose in 2}^{\text{nd}} \text{ solution}) = \frac{6.0}{100} \times 50 = 3.00 \text{ g}$ $\text{Total mass of glucose in new solution} = 1.00 + 3.00 = 4.00 \text{ g}$ $c(\text{new solution of glucose}) = \frac{4.0}{300} \times 100 = 1.3\% \text{ m/V}$
7	$600 \text{ ppm mercury} = 600 \text{ g of mercury in } 1.00 \times 10^6 \text{ g of lake water}$ $1.00 \times 10^6 \text{ g} = 1.00 \text{ Mg}$ $m(\text{Hg}) \text{ in } 5.00 \text{ Mg lake water} = 5 \times 600 = 3.00 \times 10^3 \text{ g}$ $V_2 = \frac{c_1 \times V_1}{c_2} = \frac{2.389 \times 180}{0.765} = 562 \text{ mL}$ $\therefore 382 \text{ mL } (562 - 180) \text{ of water must be added}$

Worksheet 8.3: Solutions Concentration calculations	
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8	$n(\text{K}^+ \text{ in 1 L of solution}) = c \times V = 0.0097 \times 1.00 = 0.0097 \text{ mol}$ $m(\text{K}^+ \text{ in 1 L solution}) = n \times M = 0.00970 \times 39.10 = 0.379 \text{ g}$ 1 L of sea water = 1000 mL of sea water = 1000 g of sea water $\therefore$ there is 0.379 g of K^+ in $1.00 \times 10^3 \text{ g}$ of sea water $\therefore 0.379 \times 1 \times 10^3 = 379 \text{ g of K}^+ \text{ in } 10^6 \text{ g of sea water} = 380 \text{ ppm}$
9	$c(\text{initial solution of Z}) = \frac{3.0}{2.0} = 1.50 \text{ g L}^{-1}$ $m(\text{Z in 2.0 mL of this solution}) = 1.50 \times 0.0020 = 0.00300 \text{ g}$ $\qquad\qquad\qquad = m(\text{Z in 5.0 L of dilute solution})$ $c(\text{dilute solution of Z}) = \frac{0.00300}{5.002} = 6.00 \times 10^{-4} \text{ g L}^{-1}$ $m(\text{Z in } 1.0 \mu\text{L} = 1 \times 10^{-6} \text{ L of this solution}) = 6.00 \times 10^{-4} \times 1 \times 10^{-6} = 6.0 \times 10^{-10} \text{ g}$

Worksheet 9.1: Solutions Balancing chemical equations	
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No.	Answer
1	$\text{SO}_4^{2-}(\text{aq}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Ag}_2\text{SO}_4(\text{s})$
2	$2\text{Ni}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{NiO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
3	$4\text{P}(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$
4	$2\text{CH}_3\text{COOH}(\text{aq}) + \text{K}_2\text{CO}_3(\text{s}) \rightarrow 2\text{CH}_3\text{COO}^-(\text{aq}) + 2\text{K}^+(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
5	$2\text{Al}(\text{s}) + 6\text{HI}(\text{g}) \rightarrow 2\text{AlI}_3(\text{s}) + 3\text{H}_2(\text{g})$
6	$\text{C}_5\text{H}_{12}(\text{g}) + 8\text{O}_2(\text{g}) \rightarrow 5\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
7	$\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$
8	$2\text{Cu}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{CuO}(\text{s})$
9	$2\text{KBrO}_3(\text{s}) \rightarrow 2\text{KBr}(\text{s}) + 3\text{O}_2(\text{g})$
10	$3\text{Ca}(\text{s}) + \text{N}_2(\text{g}) \rightarrow \text{Ca}_3\text{N}_2(\text{s})$
11	$2\text{H}^+(\text{aq}) + \text{Zn}(\text{OH})_2(\text{s}) \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
12	$2\text{C}_2\text{H}_4(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{H}_2\text{O}(\text{g}) + 2\text{CO}(\text{g}) + 2\text{CO}_2(\text{g})$
13	$\text{CO}_2(\text{g}) + \text{Ba}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{BaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$
14	$2\text{Rb}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Rb}^+ + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$
15	$2\text{C}_7\text{H}_{14}(\text{g}) + 21\text{O}_2(\text{g}) \rightarrow 14\text{CO}_2(\text{g}) + 14\text{H}_2\text{O}(\text{l})$
16	$4\text{Al}(\text{s}) + 3\text{O}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{Al}(\text{OH})_3(\text{s})$
17	$2\text{H}_2\text{S}(\text{g}) + \text{SO}_2(\text{g}) \rightarrow 3\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{g})$
18	$4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
19	$4\text{NH}_2\text{NHCH}_3(\text{g}) + 5\text{N}_2\text{O}_4(\text{g}) \rightarrow 9\text{N}_2(\text{g}) + 12\text{H}_2\text{O}(\text{g}) + 4\text{CO}_2(\text{g})$
20	$\text{IO}_3^-(\text{aq}) + 3\text{SO}_2(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 3\text{SO}_4^{2-}(\text{aq}) + 6\text{H}^+(\text{aq}) + \text{I}^-(\text{aq})$

Worksheet 9.2: Solutions Precipitation reactions	
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No.	Answer
1	a A white precipitate will form. b A blue precipitate will form c No precipitate will form. The contents of the test tube will remain as a colourless solution.
2	a Iron(III) sulfide b $2\text{Fe}^{3+}(\text{aq}) + 3\text{S}^{2-}(\text{aq}) \rightarrow \text{Fe}_2\text{S}_3(\text{s})$
3	a $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ (NO_3^- and Na^+ spectator ions) b $\text{Pb}^{2+}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow \text{PbI}_2(\text{s})$ (NO_3^- and Na^+ spectator ions) c $\text{Co}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Co}(\text{OH})_2(\text{s})$ (NO_3^- and K^+ spectator ions)
4	The possible products are sodium nitrate and magnesium sulfate, both of which are soluble, so a colourless solution will remain after the two reactants are mixed.
5	a Various answers are possible; for example, calcium nitrate and sodium carbonate, and calcium nitrate and lithium carbonate. One solution must contain a soluble calcium salt and the other a soluble carbonate. b $\text{Ca}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CaCO}_3(\text{s})$
6	a Any two of strontium ions, barium ions, mercury(II) ions or lead(II) ions. b Strontium and barium ions could be distinguished from mercury(II) and lead ions by adding a solution of silver nitrate. To distinguish between strontium and barium ions, a reasonable concentration of sodium hydroxide could be added and the strontium ion will form a precipitate. To distinguish between mercury(II) ions and lead ions, a dilute solution of silver nitrate could be added and the mercury(II) ions will form a precipitate, but the lead ions may not.
7	Any solution of a soluble nickel salt or chromium salt and a solution of a soluble hydroxide, carbonate or phosphate.

Worksheet 9.3: Solutions Ions in solution	
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No.	Answer
1	$2\text{H}^+(\text{aq}) + \text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
2	There are no spectator ions.
3	Mobile charged particles (ions)
4	a A b B
5	Ions present at A are H^+ and SO_4^{2-} . As the $\text{Ba}(\text{OH})_2$ solution is added, the Ba^{2+} precipitate from the solution with the SO_4^{2-} ions. The OH^- ions react with the H^+ ions, removing them from the solution as water molecules. Thus the number of ions in the solution decreases as the $\text{Ba}(\text{OH})_2$ solution is added.
6	At B, the reaction is complete and all the H^+ and SO_4^{2-} ions from the acid have been removed from the solution. With no ions present (apart from the few due to the self-ionisation of water), the conductivity is at a minimum.
7	As an excess of $\text{Ba}(\text{OH})_2$ solution (after the reaction is complete) is added, Ba^{2+} and OH^- ions are introduced into the solution. These ions are free to move in the solution and so conductivity increases.
8	$n(\text{H}_2\text{SO}_4) = c \times V = 0.15 \times 0.0200 = 0.00300 \text{ mol} = n(\text{SO}_4^{2-})$ $n(\text{Ba}^{2+}) \text{ required to react with the } \text{SO}_4^{2-} \text{ ions} = 0.00300 \text{ mol} = n(\text{Ba}(\text{OH})_2)$ $V(\text{Ba}(\text{OH})_2) = \frac{n}{c} = \frac{0.00300}{0.20} = 0.015 \text{ L} = 15 \text{ mL}$
9	No. Na_2SO_4 is soluble, so no precipitation reaction will occur. As Na^+ and OH^- ions are added, the OH^- ions react with the H^+ ions to form water, but the Na^+ ions will enter the solution and contribute to the conductivity, effectively replacing the removed H^+ . Therefore the conductivity will not decrease in the manner shown on the graph.

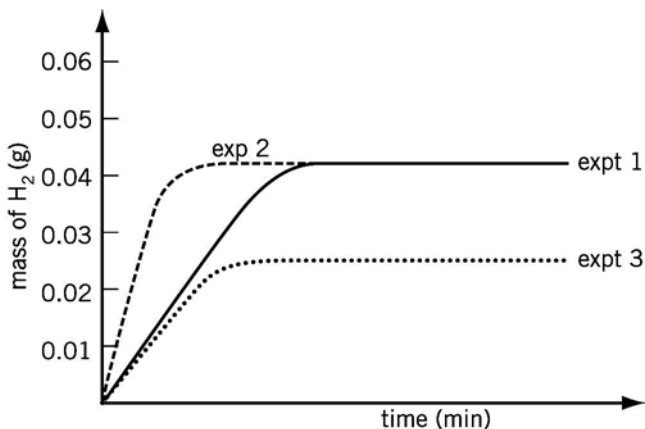
Worksheet 9.4: Solutions Introductory stoichiometry	
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No.	Answer
1	$n(\text{CH}_4) = 4.50 \text{ mol}$ $n(\text{H}_2\text{O}) = 2 \times n(\text{CH}_4) = 2 \times 4.50 = 9.00 \text{ mol}$
2	$n(\text{NaCl}) = 500 \text{ mol}$ $n(\text{Cl}_2) = \frac{1}{2} \times n(\text{NaCl}) = 250 \text{ mol}$ $\therefore m(\text{Cl}_2) = n \times M = 250 \times 70.9 = 17\,720 \text{ g} = 17.7 \text{ kg}$
3	$n(\text{Cu}(\text{OH})_2) = 0.0720 \text{ mol}$ $n(\text{OH}^-) \text{ required} = 2 \times 0.0720 = 0.1440 \text{ mol} = n(\text{NaOH})$ $V(\text{NaOH}) = \frac{n}{c} = \frac{0.1440}{0.100} = 1.44 \text{ L}$
4	$n(\text{NO}) = \frac{V}{22.41} = \frac{215}{22.41} = 9.594 \text{ mol}$ $n(\text{Cu}) = \frac{3}{2} \times n(\text{NO}) = \frac{3}{2} \times 9.594 = 14.4 \text{ mol}$
5	$n(\text{Al}) = \frac{m}{M} = \frac{5.49}{26.98} = 0.2035 \text{ mol}$ $n(\text{Cu}) = \frac{3}{2} \times n(\text{Al}) = 0.3052 \text{ mol}$ $\therefore m(\text{Cu}) = n \times M = 0.3052 \times 63.55 = 19.4 \text{ g}$
6	$n(\text{H}_3\text{PO}_4) = \frac{m}{M} = \frac{1000}{97.994} = 10.20 \text{ mol}$ $n(\text{NO}_2) = \frac{5}{1} \times n(\text{H}_3\text{PO}_4) = 5 \times 10.20 = 51.02 \text{ mol}$ $\therefore V(\text{NO}_2) = n \times 22.41 = 51.02 \times 22.41 = 1140 \text{ L}$
7	a $2\text{CH}_3\text{COOH}(\text{aq}) + \text{Na}_2\text{CO}_3(\text{s}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) + 2\text{Na}^+(\text{aq}) + 2\text{CH}_3\text{COO}^-(\text{aq})$ b $n(\text{CH}_3\text{COOH}) = c \times V = 0.156 \times 0.120 = 0.01872 \text{ mol}$ $n(\text{Na}_2\text{CO}_3) = \frac{1}{2} \times n(\text{CH}_3\text{COOH}) = 0.009360$ $m(\text{Na}_2\text{CO}_3) = n \times M = 0.009360 \times 105.99 = 0.144 \text{ g}$
8	a $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ b $n(\text{CO}_2) = 2 \times n(\text{C}_2\text{H}_6) = 2 \times 0.200 = 0.400 \text{ mol}$ c $V(\text{O}_2) = \frac{7}{2} \times V(\text{C}_2\text{H}_6) = \frac{7}{2} \times 4.30 = 15.0 \text{ L}$

Worksheet 10.1: Solutions Thermochemical equations	
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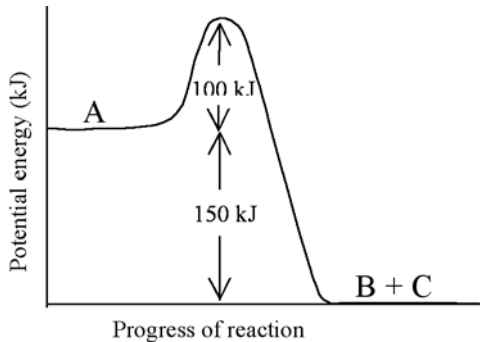
No.	Answer
1	a Endothermic b Exothermic c Endothermic d Exothermic e Exothermic f Endothermic
2	a Decrease in temperature b Increase in temperature
3	a See the endothermic example in Figure 10.5, page 232 (textbook) for a similar diagram, where the enthalpy of (S + O ₂) is greater than the enthalpy of SO ₂ . The difference in enthalpy between the products and reactant will be 297 kJ. b The energy required to break the bonds in the SO ₂ molecules will be larger than the energy released when the bonds in O ₂ and S form.
4	a $-134.8 \text{ kJ mol}^{-1}$ (2×-67.4) b $+33.7 \text{ kJ mol}^{-1}$ ($\frac{1}{2} \times +67.4$)
5	Energy released = $\frac{5}{3} \times 92.3 = 154 \text{ kJ}$
6	From the equation, 1 mol of propane produces 2220 kJ of energy, so x mol produces 1530 kJ. $\therefore x = \frac{1530}{2220} = 0.68919 \text{ mol}$ $m(\text{C}_3\text{H}_8) = n \times M = 0.68919 \times 44.094 = 30.39 \text{ g}$
7	a increase b larger c taken d exothermic e endothermic f released g endothermic h released to i larger j positive

Worksheet 11.1: Solutions Rate of reaction	
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No.	Answer
1	The particles must collide with sufficient kinetic energy to provide the required activation energy for the reaction. They must also collide with the correct orientation.
2	a $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{Na}^+(\text{aq}) + 2\text{OH}^-(\text{aq}) + \text{H}_2(\text{g})$ b Exothermic c Probably a small activation energy because the reaction between water and sodium is spontaneous. d see Figure 11.8, page 253 in the textbook for a similar diagram, even though the activation energy (shown as the peak height) could be made smaller and the heat of reaction (difference between the products and reactant energies) made larger.
3	The reaction of hydrochloric acid with magnesium is $2\text{H}^+(\text{aq}) + \text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ The higher concentration of hydrochloric acid ensures a greater concentration of H^+ ions (from the ionisation of the HCl molecules) around the magnesium metal. As a result, the rate of collision between the H^+ ions and the magnesium atom will be greater, resulting in an increase in the reaction rate.
4	a An increase in concentration of a substance increases the collision rate between the reacting particles, thus increasing the reaction rate. An increase in temperature increases the average kinetic energy of the reactant particles. As a result, more particles now have, on collision, the required activation energy to react. b In its vapour form, petrol has a maximised surface area so there is the possibility of a greater rate of collision between the petrol molecules and oxygen molecules, resulting in an increased reaction rate. That is, petrol will be more flammable in the vapour form than in the liquid phase. c Refrigeration decreases the temperature, thus decreasing the average kinetic energy of the molecules in the microorganisms. As a result, fewer molecules will have the required activation energy for a reaction to occur and the actions of the microorganisms will slow down, causing less decay.
5	

Worksheet 11.2: Solutions

The role of catalysts

No.	Answer
1	 a b E_A for the reverse reaction = 250 kJ c The activation energy peak will be less than 100 kJ, but the energies of the reactant and products must stay the same (ΔH does not change for a catalysed reaction)
2	Solid catalysts rely on reactant molecules adsorbing onto the surface of the catalyst. Pellets and wire gauzes both maximise the available surface area of the catalyst and so increase its efficiency.
3	Biological catalysts are known as enzymes.
4	Enzymes are extraordinarily efficient, but generally operate under only very mild conditions of temperature, pressure and pH, as found in the body. Any significant change from these conditions and the enzymes are denatured (cease to function). For these reasons they cannot be used in the majority of industrial processes.
5	Many catalysts function by providing a surface to which the reactant molecules can adsorb. As the molecules 'stick' to the surface of the catalyst their bond strength is decreased and the molecules break apart more readily. This has the effect of lowering the activation energy of the reaction. As the catalyst is merely providing a surface on which the reaction is initiated, it does not physically take part in the reaction and so remains unaffected. Some catalysts actually take part in the reaction, but are regenerated at some stage during the reaction back to their original form. That is, they are not consumed in the reaction but, by providing an alternative pathway in which they participate, they speed up the reaction rate.

Worksheet 12.1: Solutions Stoichiometry	
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No.	Answer
1	$n(\text{SO}_2) = \frac{V}{22.41} = \frac{50.0}{22.41} = 2.231 \text{ mol}$ $n(\text{H}_2\text{SO}_4) = n(\text{SO}_2)$ $m(\text{H}_2\text{SO}_4) = n \times M = 2.231 \times 98.076 = 219 \text{ g}$
2	$n(\text{NaHCO}_3) = c \times V = 2.86 \times 0.0299 = 0.08551 \text{ mol} = n(\text{HCO}_3^-)$ $n(\text{H}^+) \text{ required} = n(\text{HCO}_3^-) = 0.08551 \text{ mol} = n(\text{HCl})$ $c(\text{HCl}) = \frac{n}{V} = \frac{0.08551}{0.0238} = 3.59 \text{ mol L}^{-1}$
3	a $2\text{NaN}_3(\text{s}) \rightarrow 2\text{Na}(\text{s}) + 3\text{N}_2(\text{g})$ b $n(\text{NaN}_3) = \frac{m}{M} = \frac{75}{65.02} = 1.15 \text{ mol}$ $n(\text{N}_2) = \frac{3}{2} \times n(\text{NaN}_3) = \frac{3}{2} \times 1.15 = 1.73 \text{ mol}$ $V(\text{N}_2) = n(\text{N}_2) \times 22.41 = 39 \text{ L}$
4	$m(\text{Br}_2) = \text{density} \times \text{volume} = 3.12 \times 10.0 = 31.2 \text{ g}$ $n(\text{Br}_2) = \frac{m}{M} = \frac{31.2}{159.8} = 0.1952 \text{ mol}$ $n(\text{NaOH}) = \frac{10}{4} \times n(\text{Br}_2) = 0.4881 \text{ mol}$ $c(\text{NaOH}) = \frac{n}{V} = \frac{0.4881}{0.239} = 2.04 \text{ mol L}^{-1}$
5	a $n(\text{As}) = \frac{m}{M} = \frac{6.57}{74.92} = 0.08769 \text{ mol}$ $n(\text{NaOH}) = c \times V = 0.779 \times 0.250 = 0.1948 \text{ mol} = n(\text{OH}^-)$ mole ratio, in reaction of As : OH⁻ = 1 : 3, $\therefore$ 0.08769 mol of As would react with 0.2631 mol of OH⁻ $\therefore$ OH⁻ is the limiting reagent. $n(\text{H}_2) \text{ produced} = \frac{1}{2} \times n(\text{OH}^-) = \frac{1}{2} \times 0.1948 = 0.0974 \text{ mol}$ $m(\text{H}_2) = n \times M = 0.0974 \times 2.008 = 0.196 \text{ g}$ b $n(\text{AsO}_3^{2-} \text{ formed in reaction}) = \frac{1}{3} \times n(\text{OH}^-) = \frac{1}{3} \times 0.1948 = 0.06493 \text{ mol} =$ $n(\text{Na}_3\text{AsO}_3)$ $c(\text{Na}_3\text{AsO}_3) = \frac{n}{V} = \frac{0.06493}{0.250} = 0.260 \text{ mol L}^{-1}$

Worksheet 12.2: Solutions Solution stoichiometry	
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No.	Answer
1	$n(\text{Ba}(\text{NO}_3)_2) = c \times V = 0.450 \times 0.2000 = 0.0900 \text{ mol} = n(\text{Ba}^{2+})$ $n(\text{BaSO}_4) = n(\text{Ba}^{2+}) = 0.0900 \text{ mol}$
2	$n(\text{H}_2\text{O}_2) = c \times V = 4.30 \times 3.50 = 15.05 \text{ mol}$ $n(\text{gaseous product}) = \frac{5}{2} \times n(\text{H}_2\text{O}_2) = \frac{5}{2} \times 15.05 = 37.62 \text{ mol}$ $V(\text{gases}) = n(\text{gases}) \times 22.41 = 37.62 \times 22.41 = 843 \text{ L}$
3	$n(\text{SO}_2) = \frac{m}{M} = \frac{70.9}{64.06} = 1.107 \text{ mol}$ $n(\text{OH}^-) = 2 \times n(\text{SO}_2) = 2.214 \text{ mol} = n(\text{NaOH})$ $\therefore V(\text{NaOH}) = \frac{n}{c} = \frac{2.214}{0.638} = 3.47 \text{ L}$
4	$n(\text{HNO}_3) = c \times V = 14.0 \times 0.125 = 1.75 \text{ mol}$ $n(\text{Hg}) = \frac{3}{8} \times n(\text{HNO}_3) = 0.6562 \text{ mol}$ $\therefore m(\text{Hg}) = n \times M = 0.6562 \times 200.6 = 132 \text{ g}$
5	$n(\text{Hg}(\text{NO}_3)_2) = c \times V = 0.144 \times 0.0336 = 4.838 \times 10^{-3} \text{ mol} = n(\text{Hg}^{2+})$ $n(\text{Br}^-) = \frac{2}{1} \times n(\text{Hg}^{2+}) = 9.677 \times 10^{-3} \text{ mol} = n(\text{NH}_4\text{Br})$ $\therefore V(\text{NH}_4\text{Br}) = \frac{n}{c} = \frac{0.009677}{0.0712} = 0.136 \text{ L}$
6	$n(\text{HCl}) = c \times V = 0.500 \times 0.01885 = 0.009425 \text{ mol} = n(\text{H}^+)$ $n(\text{Na}_2\text{CO}_3) = \frac{1}{2} \times n(\text{H}^+) = 0.004712 \text{ mol}$ $m(\text{Na}_2\text{CO}_3) = n \times M = 0.004712 \times 105.99 = 0.4994 \text{ g}$ $\% \text{ purity of Na}_2\text{CO}_3 \text{ sample} = \frac{0.4994}{0.600} \times 100 = 83.2\%$
7	$m(\text{CH}_3\text{COOH in vinegar sample}) = \frac{4.58}{100} \times 14.72 = 0.6742 \text{ g}$ $n(\text{CH}_3\text{COOH}) = \frac{m}{M} = \frac{0.6742}{60.052} = 0.01123 \text{ mol}$ $n(\text{NaOH}) = c \times V = 0.307 \times 0.02500 = 0.007675 \text{ mol} = n(\text{OH}^-)$ Reacting ratio of $\text{CH}_3\text{COOH} : \text{OH}^- = 1:1$, $\therefore \text{OH}^-$ is the limiting reagent $n(\text{CH}_3\text{COO}^- \text{ formed}) = n(\text{OH}^- \text{ reacted}) = 0.007675$ $c(\text{CH}_3\text{COO}^-) = \frac{n}{V} = \frac{0.007675}{(0.100 + 0.02500)} = 0.0614 \text{ mol L}^{-1}$

Worksheet 12.2: Solutions Solution stoichiometry	
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Worksheet 12.3: Solutions Excess reagent stoichiometry	
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No.	Answer
1	a $\text{C}_2\text{H}_4(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ b From the equation, 1 mol of ethene reacts with 3 mol of oxygen. Therefore, 2.0 mol of ethene will react with 6.0 mol of oxygen so there are 3 mol of oxygen gas in excess, and the ethene is the limiting reagent. c 2.0 mol of ethene will form 4.0 mol of carbon dioxide
2	$m(\text{H}_2\text{O}) = \text{density} \times \text{volume} = 1.00 \times 180 = 180 \text{ g}$ $\therefore n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{180}{18.016} = 9.991 \text{ mol}$ $n(\text{Na}) = \frac{m}{M} = \frac{0.835}{22.99} = 0.03632 \text{ mol}$ 0.03632 mol of Na will react with 0.03632 mol of H_2O $\therefore$ Na is the limiting reagent $n(\text{NaOH}) = n(\text{Na}) = 0.03632 \text{ mol}$ $\therefore c(\text{NaOH}) = \frac{n}{V} = \frac{0.03632}{0.180} = 0.202 \text{ mol L}^{-1}$ (assuming the amount of water reacting is very small and has not changed the volume of water present. In reality, $0.03632 \times 18.016 = 0.654 \text{ g} = 0.654 \text{ mL}$ of water will react, so the volume will change to 0.1793 L, so the $c(\text{NaOH})$ will be 0.203 mol L^{-1})
3	$n(\text{Cr}_2\text{O}_3) = \frac{m}{M} = \frac{10.0}{152.0} = 0.06579 \text{ mol}$ $n(\text{KBrO}_3) = c \times V = 0.774 \times 0.120 = 0.09288 \text{ mol} = n(\text{BrO}_3^-)$ $n(\text{BrO}_3^-)$ consumed in reaction $= \frac{6}{5} \times n(\text{Cr}_2\text{O}_3) = 0.07895 \text{ mol}$ $\therefore$ KBrO_3 is in excess and the Cr_2O_3 is the limiting reagent $n(\text{Br}_2)$ formed $= \frac{3}{5} \times n(\text{Cr}_2\text{O}_3) = 0.03947 \text{ mol}$ $\therefore m(\text{Br}_2) = n \times M = 0.03947 \times 159.8 = 6.31 \text{ g}$
4	a $\text{Sn}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Sn}(\text{OH})_2(\text{s})$ b $n(\text{SnCl}_2) = c \times V = 0.968 \times 0.0455 = 0.04404 \text{ mol} = n(\text{Sn}^{2+})$ $n(\text{NH}_4\text{OH}) = c \times V = 1.223 \times 0.0582 = 0.07118 \text{ mol} = n(\text{OH}^-)$ $n(\text{Sn}^{2+})$ consumed in reaction $= \frac{1}{2} \times n(\text{OH}^-) = 0.03558 \text{ mol}$ $\therefore$ Sn^{2+} is in excess and OH^- is the limiting reagent $n(\text{Sn}(\text{OH})_2)$ precipitated $= \frac{1}{2} \times n(\text{OH}^-) = 0.03558 \text{ mol}$ $\therefore m(\text{Sn}(\text{OH})_2) = n \times M = 0.03558 \times 152.716 = 5.43 \text{ g}$

Worksheet 12.4: Solutions Empirical and molecular formula calculations	
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No.	Answer
1	$\% \text{ Ti} = \frac{M(\text{Ti})}{M(\text{TiO}_2)} \times \frac{100}{1} = \frac{47.88}{79.88} \times \frac{100}{1} = 59.9\%$ $\% \text{ O} = \frac{2 \times M(\text{O})}{M(\text{TiO}_2)} \times \frac{100}{1} = \frac{32.00}{79.88} \times \frac{100}{1} = 40.1\% \text{ (or simply } 100 - 59.9 = 40.1)$
2	$\text{mol ratio S : O} = \frac{40.0}{32.06} : \frac{60.0}{16.00} = 1.248 : 3.75 = 1 : 3 \therefore \text{SO}_3$
3	Let the unknown metal be X. As it is in group 1 the empirical formula must be XCl. The percentage composition of X must be 52.4% $\text{mole ratio } \frac{52.4}{M(\text{X})} : \frac{47.6}{35.45} = 1 : 1 \therefore \frac{52.4}{M(\text{X})} = \frac{47.6}{35.45} \therefore M(\text{X}) = \frac{52.4 \times 35.45}{47.6} = 39.02$ That is, the atomic mass of X is approximately 39.02 and so X must be K The formula of its chloride is KCl.
4	a CH₄ b NH₂ c CH₂O
5	$M_r(\text{compound}) = 88, M_r(\text{C}_2\text{H}_4\text{O}) = 44$, so the ratio is $\frac{88}{44} = 2$ The molecular formula is $2 \times (\text{C}_2\text{H}_4\text{O}) = \text{C}_4\text{H}_8\text{O}_2$
6	$M(\text{hydrocarbon}) = \frac{m}{n} = \frac{24.6}{0.30} = 82 \text{ g mol}^{-1}$ $M_r(\text{C}_3\text{H}_5) = 41 = (82 \div 2)$, so the molecular formula is $2 \times (\text{C}_3\text{H}_5) = \text{C}_6\text{H}_{10}$
7	$\text{K : Mn : O} = \frac{0.614}{39.10} : \frac{0.863}{54.94} : \frac{1.006}{16.00}$ $= 0.01570 : 0.01571 : 0.06288$ $= 1 : 1 : 4$ Hence the empirical formula is KMnO₄
8	$\text{C : H : O} = \frac{40.0}{12.01} : \frac{6.67}{1.008} : \frac{53.3}{16.00}$ $= 3.331 : 6.617 : 3.331$ $= 1 : 1.99 : 1 \approx 1 : 2 : 1 \text{ hence the empirical formula is CH}_2\text{O}$ $M_r(\text{CH}_2\text{O}) = 30 = (60 \div 2)$, so the molecular formula is $2 \times (\text{CH}_2\text{O}) = \text{C}_2\text{H}_4\text{O}_2$

Worksheet 12.5: Solutions Determining the molecular formula of a gas	
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No	Answer
1	$\begin{array}{ccc} \text{Na} & : & \text{B} & : & \text{O} \\ \hline 22.8 & : & 21.5 & : & 55.7 \\ 22.99 & : & 10.81 & : & 16.00 \\ 0.9917 & : & 1.989 & : & 3.481 \\ 1 & : & 2.006 & : & 3.510 \\ 2 & : & 4 & : & 7 \end{array}$ Empirical formula is $\text{Na}_2\text{B}_4\text{O}_7$.
2	$\begin{array}{ccc} \text{C} & : & \text{H} & : & \text{O} \\ \hline 0.0556 & : & \frac{0.112}{1.008} & : & \frac{9.57 \times 10^{21}}{6.022 \times 10^{23}} \\ 0.0556 & : & 0.1111 & : & 0.01589 \\ 3.499 & : & 6.992 & : & 1 \\ (\times 2) & 6.998 & : & 13.98 & : & 2 \approx 7 : 14 : 2 \end{array}$ Empirical formula is $\text{C}_7\text{H}_{14}\text{O}_2$.
3	$\begin{array}{ccc} \text{As} & : & \text{O} \\ \hline 75.7 & : & 24.3 \\ 74.92 & : & 16.00 \\ 1.010 & : & 1.519 \\ (\times 2) & 2.02 & : & 3.038 \approx 2 : 3 \end{array}$ Empirical formula is As_2O_3. Formula mass = 197.84 MF is $(\text{As}_2\text{O}_3)_a$, where $a = \frac{\text{molecular mass}}{\text{formula mass}} = \frac{395.6}{197.84} \approx 2$ Molecular formula is As_4O_6.
4	It absorbs water formed in the reaction.
5	It absorbs carbon dioxide formed in the reaction.

Worksheet 12.5: Solutions Determining the molecular formula of a gas	
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No	Answer
6	$m(\text{CO}_2 \text{ formed}) = 43.54 \text{ g}$ $n(\text{CO}_2) = \frac{43.54}{44.01} = 0.9893 = n(\text{C in hydrocarbon})$ $m(\text{H}_2\text{O formed}) = 22.26 \text{ g}$ $n(\text{H}_2\text{O}) = \frac{22.26}{18.016} = 1.236$ $n(\text{H in hydrocarbon}) = 1.236 \times 2 = 2.471$ C : H Mole ratio 0.9893 : 2.471 1 : 2.498 (×2) 2 : 4.996 $\approx$ 2 : 5 Empirical formula is C_2H_5.
7	The sample must vaporise at a temperature below that of the steam (below 100°C).
8	$PV = 8.315 \text{ nT}$ $n(\text{gas formed from liquid}) = \frac{101.3 \times 0.046}{8.315 \times (100 + 273.1)} = 0.001502 \text{ mol}$ $M(\text{liquid}) = \frac{m}{n} = \frac{0.16}{0.001502} = 107 \text{ g mol}^{-1}$
9	$m(\text{C}) \text{ in sample} = \frac{12.0}{44.0} \times 27.83 = 7.590 \text{ g}$ $m(\text{H}) \text{ in sample} = \frac{2.0}{18.0} \times 11.38 = 1.264 \text{ g}$ $m(\text{O}) \text{ in sample} = 18.99 - 7.590 - 1.264 = 10.14 \text{ g}$ C : H : O Mole ratio $\frac{7.590}{12.0}$: $\frac{1.264}{1.0}$: $\frac{10.14}{16.0}$ 0.633 : 1.264 : 0.634 1 : 2 : 1 Empirical formula is CH_2O. $M = \frac{8.31mT}{pV} = \frac{6.21 \times 8.31 \times 473}{1.25 \times 10^2 \times 2.17} = 89.99 \text{ g mol}^{-1}$ EFM = 30 MF is $(\text{CH}_2\text{O})_a$, where $a = \frac{\text{RMM}}{\text{EFM}} = \frac{90}{30} = 3$ Molecular formula is $\text{C}_3\text{H}_6\text{O}_3$.

Worksheet 12.6: Solutions Gravimetric analysis of a fertiliser	
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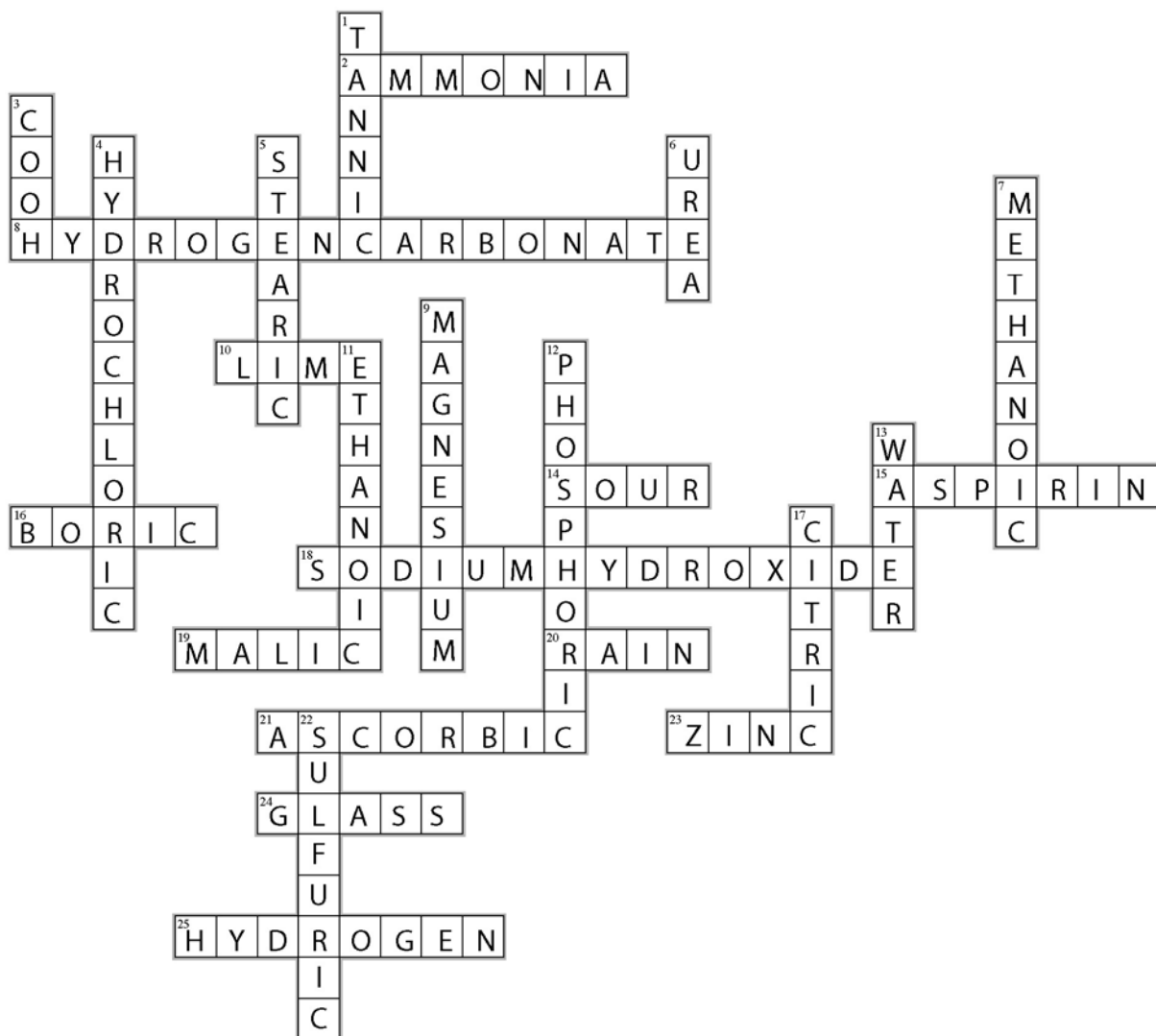
No.	Answer
1	It was necessary to dry it so that water did not add any mass to the compound. Grinding it helped it to dissolve.
2	$n((\text{NH}_4)_2\text{SO}_4) = \frac{m}{M} = \frac{1.00}{132.144} = 7.57 \times 10^{-3} \text{ mol}$
3	Between steps 5 and 6, additional filtration with washing of the residue would have been required to remove the insoluble components.
4	$\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ If the fertiliser was pure: $n(\text{BaCl}_2) \text{ needed} = n(\text{Ba}^{2+}) = n(\text{SO}_4^{2-}) = n((\text{NH}_4)_2\text{SO}_4) = 7.567 \times 10^{-3} \text{ mol}$ $V(\text{BaCl}_2) = \frac{n}{c} = \frac{7.567 \times 10^{-3}}{0.200} = 0.0378 \text{ L} = 37.8 \text{ mL}$
5	The precipitating agent should be in excess to ensure complete precipitation.
6	The supernatant liquid (filtrate) was tested with BaCl_2 solution. No precipitate was observed.
7	Heat helps larger crystals of BaSO_4 form so it is better to wash it in hot water than cold. Washing removes any ions adsorbed onto the surface of the precipitate.
8	This means that the precipitate was dried, weighed, then dried and weighed again. If the masses are the same, then constant mass has been achieved and there is no more water present.
9	$n(\text{BaSO}_4) = \frac{m}{M} = \frac{1.63}{233.36} = 6.985 \times 10^{-3} \text{ mol} = n(\text{SO}_4^{2-}) = n((\text{NH}_4)_2\text{SO}_4)$ $m((\text{NH}_4)_2\text{SO}_4) = n \times M = 6.985 \times 10^{-3} \times 132.144 = 0.9230 \text{ g}$ $\% \text{ m/m } ((\text{NH}_4)_2\text{SO}_4) = \frac{0.9230}{1.00} \times 100 = 92.3\%$
10	• Errors in weighing • Incomplete dissolving of fertiliser • Incomplete precipitation • Losses due to filtration • Not completely drying the precipitate

Worksheet 13.1: Solutions

Common acids and bases

NAME:

CLASS:



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Worksheet 13.1: Solutions

Common acids and bases

ACROSS

2. AMMONIA—Many window cleaners contain this basic substance.
8. HYDROGENCARBONATE—The sodium salt of this ion is baking soda.
10. LIME—A common name for a hydroxide used to neutralise soil acidity.
14. SOUR—The taste of an acid.
15. ASPIRIN—The analgesic chemically known as acetyl salicylic.
16. BORIC—_____ acid is a mild antiseptic used in ointments for dressing wounds.
18. SODIUMHYDROXIDE—A caustic chemical used in drain cleaners. (6, 9)
19. MALIC—_____ acid is found in grapes and apples, particularly unripe ones.
20. RAIN—Carbonic acid is found naturally in this watery substance.
21. ASCORBIC—_____ acid is Vitamin C.
23. ZINC—Many watch batteries have this hydroxide as a reactant.
24. GLASS—Hydrofluoric acid is used to etch this substance.
25. HYDROGEN—Produced when hydrochloric acid reacts with iron.

DOWN

1. TANNIC—_____ acid is found in wine and tea.
3. COOH—The chemical formula for the carboxylic acid functional group.
4. HYDROCHLORIC—_____ acid is the main component of concrete cleaner.
5. STEARIC—A fatty acid with molecular formula of $C_{18}H_{36}O_2$, found widely in solid fats.
6. UREA—Human urine contains this basic substance.
7. METHANOIC—_____ acid is the systematic name for the acid present in bee stings.
9. MAGNESIUM—One of the hydroxides commonly used in antacid preparations.
11. ETHANOIC—_____ acid is the systematic name for the acid in vinegar.
12. PHOSPHORIC—_____ acid is an active ingredient in Coca-Cola.
13. WATER—One of the products of the reaction between an acid and a metal hydroxide.
17. CITRIC—Lemons and oranges contain _____ acid.
22. SULFURIC—_____ acid is found in most car batteries.

Worksheet 13.2: Solutions Revision of acids and bases	
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No.	Answer
1	a $\text{KHCO}_3(\text{s}) + \text{H}^+(\text{aq}) \rightarrow \text{K}^+(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$ b $\text{Fe}_2\text{O}_3(\text{s}) + 6\text{H}^+(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$ c $\text{OH}^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
2	a $\text{Ca}(\text{OH})_2(\text{s}) \xrightarrow{\text{H}_2\text{O}} \text{Ca}_2^+(\text{aq}) + 2\text{OH}^-(\text{aq})$ b $\text{H}_2\text{SO}_4(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$ or $\text{H}_2\text{SO}_4(\text{l}) \xrightarrow{\text{H}_2\text{O}} \text{H}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$ $\text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ or $\text{HSO}_4^-(\text{aq}) \xrightarrow{\text{H}_2\text{O}} \text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ c $\text{OH}^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
3	a S^{2-} and O^{2-} b H_2SO_4 and H_3PO_4
4	a $\text{HCO}_3^-/\text{CO}_3^{2-}$ and $\text{H}_3\text{O}^+/\text{H}_2\text{O}$ b $\text{HS}^-/\text{S}^{2-}$ and $\text{H}_2\text{O}/\text{OH}^-$ c HCl/Cl^- and $\text{NH}_4^+/\text{NH}_3$
5	a $\text{HSO}_4^-(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ b $\text{HSO}_4^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) \rightarrow \text{H}_2\text{SO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
6	Different acid strengths (e.g. one strong, one weak) or different acid types (e.g. one monoprotic, one diprotic)
7	a Methanoic acid is a weak acid that only partially ionises in water, producing few ions and so having low conductivity. Hydrochloric acid is a strong acid that ionises completely in water, producing many ions and so having high conductivity. b Only the O–H bond has sufficient polarity to make the hydrogen atom acidic. The three C–H bonds are only slightly polar, so these hydrogen atoms are not acidic.
8	The solutions listed in order of decreasing pH are: NaOH is a strong base, $\therefore$ high pH NH_3 is a weak base, $\therefore$ pH > 7 but not too high H_2O has pH = 7 CH_3COOH is a weak acid, $\therefore$ pH < 7 HNO_3 is a strong acid, $\therefore$ low pH H_2SO_4 is a strong, diprotic acid, $\therefore$ pH < pH of HNO_3
9	A 10-fold dilution produces a change of one pH unit so the pH must increase by 1 unit to a value of 4.
10	a $\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{CO}_3(\text{aq})$ b $\text{SO}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{SO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ c no reaction d $\text{Al}_2\text{O}_3(\text{s}) + 2\text{OH}^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Al}(\text{OH})_4^-(\text{aq})$

Worksheet 13.2: Solutions Revision of acids and bases	
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Worksheet 14.1: Solutions Oxidation numbers and redox equations	
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No.	Answer
1	Examples: a NF ₃ b N ₂ O ₅ c N ₂ d NH ₃
2	a +6 b +6 c +4 d -1
3	a Not redox b Oxidant = I ₂ O ₅ , reductant = CO c Not redox d Oxidant = Hg ²⁺ , reductant = N ₂ H ₄ e Oxidant = NO ₃ ⁻ , reductant = H ₂ S f Oxidant = NO ₂ , reductant = NO ₂
4	a +5 b +2 c +7
5	C = 0 H = +1 O = -2
6	a Oxidation: CH ₃ CH ₂ OH(aq) + H ₂ O(l) → CH ₃ COOH(aq) + 4H ⁺ (aq) + 4e ⁻ Reduction: Cr ₂ O ₇ ²⁻ (aq) + 14H ⁺ (aq) + 6e ⁻ → 2Cr ³⁺ (aq) + 7H ₂ O(l) Redox: 3CH ₃ CH ₂ OH(aq) + 2Cr ₂ O ₇ ²⁻ (aq) + 16H ⁺ (aq) → 3CH ₃ COOH(aq) + 4Cr ³⁺ (aq) + 11H ₂ O(l) b Oxidation: 2I ⁻ (aq) → I ₂ (aq) + 2e ⁻ Reduction: BrO ₃ ⁻ (aq) + 6H ⁺ (aq) + 6e ⁻ → Br ⁻ (aq) + 3H ₂ O(l) Redox: 6I ⁻ (aq) + BrO ₃ ⁻ (aq) + 6H ⁺ (aq) → 3I ₂ (aq) + Br ⁻ (aq) + 3H ₂ O(l) c Oxidation: Fe ²⁺ (aq) → Fe ³⁺ (aq) + e ⁻ Reduction: MnO ₄ ⁻ (aq) + 8H ⁺ (aq) + 5e ⁻ → Mn ²⁺ (aq) + 4H ₂ O(l) Redox: 5Fe ²⁺ (aq) + MnO ₄ ⁻ (aq) + 8H ⁺ (aq) → 5Fe ³⁺ (aq) + Mn ²⁺ (aq) + 4H ₂ O(l) d Oxidation: H ₂ O ₂ (aq) → O ₂ (g) + 2H ⁺ (aq) + 2e ⁻ Reduction: MnO ₄ ⁻ (aq) + 8H ⁺ (aq) + 5e ⁻ → Mn ²⁺ (aq) + 4H ₂ O(l) Redox: 5H ₂ O ₂ (aq) + 2MnO ₄ ⁻ (aq) + 6H ⁺ (aq) → 5O ₂ (g) + 2Mn ²⁺ (aq) + 8H ₂ O(l) e Oxidation: H ₂ S(g) → S(s) + 2H ⁺ (aq) + 2e ⁻ Reduction: Cr ₂ O ₇ ²⁻ (aq) + 14H ⁺ (aq) + 6e ⁻ → 2Cr ³⁺ (aq) + 7H ₂ O(l) Redox: 3H ₂ S(g) + Cr ₂ O ₇ ²⁻ (aq) + 8H ⁺ (aq) → 3S(s) + 2Cr ³⁺ (aq) + 7H ₂ O(l)

Worksheet 14.2: Solutions Metal reactivity and redox equations	
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No.	Answer
1	a Anything below aluminium in the list; for example, copper and lead. b Anything above calcium in the list; for example, lithium and potassium.
2	Any two of: copper, lead and silver.
3	The manganese would displace the lead(II) ions from the solution and a coating of lead (a grey solid) would form on the surface of the manganese: $\text{Mn(s)} + \text{Pb}^{2+}(\text{aq}) \rightarrow \text{Pb(s)} + \text{Mn}^{2+}(\text{aq})$
4	Nothing, because zinc is more reactive (a stronger reductant) than nickel.
5	a $\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^{-}$ $\text{Fe}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Fe(s)}$ b $\text{Mg(s)} + \text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe(s)} + \text{Mg}^{2+}(\text{aq})$
6	A more reactive metal, such as zinc, must be added. If zinc were to be added, the following reaction would occur: $\text{Zn(s)} + \text{Pb}^{2+}(\text{aq}) \rightarrow \text{Pb(s)} + \text{Zn}^{2+}(\text{aq})$
7	The nickel would react with the silver ions: $\text{Ni(s)} + 2\text{Ag}^{+}(\text{aq}) \rightarrow \text{Ni}^{2+}(\text{aq}) + 2\text{Ag(s)}$
8	The aluminium would react with the zinc nitrate, so would not be appropriate. The copper, nickel and silver would not react. Silver is the least reactive and so would be the best (but expensive!) choice.
9	Place a small piece of manganese (or aluminium or magnesium) in each. The zinc solution will react to deposit zinc. The magnesium solution will not.
10	Adding the metal to its own metal salt does not provide useful data.
11	D
12	B (group 2)
13	A and C
14	a $3\text{B(s)} + 2\text{D}^{3+}(\text{aq}) \rightarrow 2\text{D(s)} + 3\text{B}^{2+}(\text{aq})$ b $2\text{C(s)} + \text{B}^{2+}(\text{aq}) \rightarrow \text{B(s)} + 2\text{C}^{+}(\text{aq})$ c $\text{D(s)} + 3\text{A}^{+}(\text{aq}) \rightarrow 3\text{A(s)} + \text{D}^{3+}(\text{aq})$
15	C, B, D, A

Worksheet 14.3: Solutions Corrosion of iron	
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No.	Answer										
1	a $\text{Fe(s)} \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{e}^{-}$ b $\text{O}_2(\text{g}) + 2\text{H}_2\text{O(l)} + 4\text{e}^{-} \rightarrow 4\text{OH}^{-}(\text{aq})$										
2	The dessicant (anhydrous calcium chloride) removes water. With no water present, corrosion cannot occur.										
3	Boiling removes oxygen from the water. The oil prevents any further oxygen dissolution. With no oxygen present, corrosion cannot occur.										
4	Pure oxygen, rather than air, means a higher concentration of the reactant oxygen. Reaction rate increases with increasing reactant concentration.										
5	a No oxygen present. b There is very little dissolved oxygen in the deepest parts of the ocean. c The oil prevents contact between the oxygen and water in the air, and the iron surface. d The magnesium is more reactive (stronger reductant) than iron, so the magnesium, rather than the iron, will supply the electrons for the reduction of oxygen. e The negative terminal supplies electrons to the surface of the iron. With an excess of electrons present, oxidation of the iron is prevented.										
6	a The zinc acts as a mechanical barrier, preventing contact between the iron and the oxygen/water in the air. Zinc can also act as a sacrificial anode, being oxidised in preference to the iron. b The tin acts as a mechanical barrier, preventing contact between the iron and the oxygen/water in the air. c Zinc may react with foods and may be toxic.										
7	<table style="width: 100%; border: none;"> <tr> <td style="width: 50%; border: none;">Gardening secateurs</td><td style="width: 50%; border: none;">Coating with oil</td></tr> <tr> <td style="border: none;">Natural gas pipeline</td><td style="border: none;">Attachment to sacrificial anodic magnesium blocks</td></tr> <tr> <td style="border: none;">Wharf</td><td style="border: none;">Cathodic protection with a DC power supply</td></tr> <tr> <td style="border: none;">Patio furniture</td><td style="border: none;">Painting</td></tr> <tr> <td style="border: none;">A shed roof</td><td style="border: none;">Galvanising</td></tr> </table>	Gardening secateurs	Coating with oil	Natural gas pipeline	Attachment to sacrificial anodic magnesium blocks	Wharf	Cathodic protection with a DC power supply	Patio furniture	Painting	A shed roof	Galvanising
Gardening secateurs	Coating with oil										
Natural gas pipeline	Attachment to sacrificial anodic magnesium blocks										
Wharf	Cathodic protection with a DC power supply										
Patio furniture	Painting										
A shed roof	Galvanising										

Worksheet 15.1: Solutions Electrolysis reactions	
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No.	Answer
1	These elements are more reactive than water (stronger reductants) so require electrolysis of molten salts (or a reaction of the salt with a more reactive metal) to produce them. Electrolysis requires the passage of an electric current. This was not possible until Volta's invention of the first effective galvanic cell in 1800.
2	Water is able to undergo both oxidation and reduction in electrolytic cells and so may react preferentially to the cations and anions present from the salt.
3	A similar diagram to the one shown on page 361 of the textbook should be drawn. a Li^+ reacts at the negative electrode, F^- at the positive electrode. b The negative electrode is the cathode, the positive electrode is the anode. c $\text{Li}^+(\text{l}) + \text{e}^- \rightarrow \text{Li}(\text{l})$ is the reduction reaction; $2\text{F}^-(\text{l}) \rightarrow \text{F}_2(\text{g}) + 2\text{e}^-$ is the oxidation reaction. d Li^+ is the oxidant, F^- is the reductant. e $2\text{Li}^+(\text{l}) + 2\text{F}^-(\text{l}) \rightarrow 2\text{Li}(\text{l}) + \text{F}_2(\text{g})$
4	Magnesium metal would be formed at the cathode. Because sodium is more reactive (a stronger reductant) than magnesium, it can be inferred that Na^+ would be a weaker oxidant than Mg^{2+} , so Mg^{2+} will 'want' to gain electrons more readily than Na^+ . Iodine would form at the anode. Because chlorine is more reactive (a stronger oxidant) than iodine, it can be inferred that Cl^- would be a weaker reductant than I^- , so I^- will 'want' to lose electrons more readily than Cl^- .
5	a copper metal b $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ c water d $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ e oxygen
6	Lead metal would form at the cathode. Oxygen gas (and H^+ ions) would form at the anode. Oxidation reaction: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ Reduction reaction: $\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$
7	a Potassium chloride is used. (Iodine would form at the anode in the alternative case and iodine is a dark coloured solid, producing a brown colour in solution.) b Oxygen gas forms as a result of the oxidation of water. Some chlorine gas may also be present, dependent upon the concentration of the solution. c At the cathode, hydrogen gas will be evolved due to the reduction of water. d Anode reaction: $2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$ Cathode reaction: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$
8	a At the cathode: hydrogen will form as a result of the reduction of water. At the anode: the copper electrode will undergo oxidation to Cu^{2+} ions. b Cathode: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ Anode: $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

Worksheet 15.2: Solutions Redox chemistry definitions	
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No.	Answer
1	Oxidation Involves the Loss of electrons. Reduction Involves the Gain of electrons.
2	a Any change in oxidation numbers of atoms in a chemical reaction indicates that a redox reaction has occurred. b If the oxidation number of an atom in a species increases, then that species has been oxidised; if it decreases then the species has been reduced. c As the oxidation number of sulfur has decreased from +6 to -2 it has undergone reduction. This means that sulfuric acid is acting as an oxidant.
3	An oxidant species oxidises another chemical in a redox reaction. To do this, it allows the other species to lose electrons by gaining those electrons itself. The process of gaining electrons is defined as reduction; hence, the oxidant has undergone reduction.
4	Conjugate redox pairs of species differ by one or more electrons. The pairs of species that meet this criterion are A and D.
5	a An electrode provides a solid conductive material to which conducting wires can be connected to construct an electrical circuit. Electrodes are the sites of oxidation and reduction reactions. b An inert electrode does not react (that is, does not undergo oxidation) during the electrolysis process. Inert electrodes are used when it is not desirable for the electrode to react during electrolysis (and when the species undergoing oxidation is one of the electrolytes (in the molten or aqueous mixture) in the electrolysis cell). c Platinum and graphite are commonly used as inert electrodes.
6	In electrolysis cells, the negative electrode has an excess of electrons (because it is connected to the negative terminal of the external power source) and so is the site at which reduction (gain of electrons) is occurring. Reduction always occurs at the cathode.
7	The electrode connected to the positive terminal of the external power source becomes the anode because the reaction that occurs at this electrode involves the loss of electrons (oxidation) by the species in the cell (these electrons move back to the positive terminal of the power source). Oxidation always occurs at the anode.
8	In the electrolytic mixture, the cations (positive ions) flow towards the cathode (negative electrode) and the anions (negative ions) flow towards the anode (positive electrode).

Worksheet 15.2: Solutions

Redox chemistry definitions

No.	Answer
9	a A cell like the one shown on page 361 of the text book should be drawn. b The electrode that the electrons flow into is the negative electrode. At this electrode, reduction takes place. The oxidation reaction takes place at the positive electrode. c The electrons flow in the external circuit. They flow into the negative electrode of the cell and away from the positive electrode of the cell. (The negative electrode is connected to the negative terminal of the external power source and the positive electrode is connected to the positive terminal of the external power source.) In the electrolyte of the cell, the positive ions (A^{2+}) flow towards the negative electrode and the negative ions (B^{2-}) flow towards the positive electrode.

Worksheet 15.3: Solutions Recovering silver from solution	
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No.	Answer
1	aliquot – an accurately measured volume of liquid normally delivered from a pipette pipette – equipment used to accurately measure a specific volume of a liquid titrated – when a solution is added to a measured volume of a second solution in a container until the reaction is complete concordant – two or more results are concordant when they are very close in value to one another (for example, two volumes within 0.1 mL of each other could be called concordant results). The expected concordant range is dependent on the accuracy of the equipment being used. titre – a measured volume of solution added, normally from a burette, to a second solution for a complete reaction to occur.
2	a $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ b $2\text{Ag}^+(\text{aq}) + \text{CrO}_4^{2-}(\text{aq}) \rightarrow \text{Ag}_2\text{CrO}_4(\text{s})$
3	$n(\text{NaCl}) = c \times V = 0.100 \times 0.02000 = 0.002000 \text{ mol} = n(\text{Cl}^-)$ $n(\text{Ag}^+) = n(\text{Cl}^-) = 0.002000 \text{ mol} = n(\text{AgNO}_3)$ $c(\text{AgNO}_3) = \frac{n}{V} = \frac{0.002000}{0.01932} = 0.104 \text{ mol L}^{-1}$
4	There is uncertainty in the pipette and burette volumes, and in the judging of the endpoint (completion of the reaction).
5	Dry and weigh the sample until the mass is constant.
6	a $2\text{Ag}^+(\text{aq}) + \text{Cu}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Cu}^{2+}(\text{aq})$ b Silver ion
7	$n(\text{AgNO}_3) = c \times V = 0.104 \times 0.02000 = 0.002080 \text{ mol} = n(\text{Ag}^+)$ $n(\text{Ag}) = n(\text{Ag}^+) = 0.002080 \text{ mol}$ $m(\text{Ag}) \text{ present} = n \times M = 0.002070 \times 107.9 = 0.2244 \text{ g}$ $\% \text{ recovery} = \frac{\text{mass recovered}}{\text{mass present}} \times 100 = \frac{0.206}{0.2244} \times 100 = 91.8\%$
8	- Reaction between the Ag^+ ions in solution and the Cu spiral was incomplete. - Some silver remained adhering to the copper spiral. - Small amounts of silver were lost in the transfer and filtering process.
9	$\text{S}_2\text{O}_4^{2-}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{SO}_3^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^-$
10	$2\text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^-$

Worksheet 15.4: Solutions The chloro-alkali industry	
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No	Answer
1	a $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ b $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$ c $2\text{Cl}^-(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) + 2\text{OH}^-(\text{aq})$
2	$n(\text{NaCl}) = \frac{m}{M} = \frac{1.0 \times 10^6}{58.44} = 1.71 \times 10^4 \text{ mol} = n(\text{Cl}^-)$ $n(\text{Cl}_2) = \frac{1}{2} \times n(\text{Cl}^-) = \frac{1}{2} \times 1.71 \times 10^4 = 8.56 \times 10^3 \text{ mol}$ $V(\text{Cl}_2) = n(\text{Cl}_2) \times 22.41 = 8.56 \times 10^3 \times 22.41 = 1.9 \times 10^5 \text{ L}$
3	a $\text{O}_2(\text{g})$ and $\text{H}^+(\text{aq})$ b At concentrations of much greater than 1 mol L^{-1} and temperatures well above 25°C , predictions from the reactivity series often given in books are not always accurate.
4	It is easier to reduce water than Na^+ (water is a stronger oxidant than Na^+), so sodium metal will never be produced in aqueous solution.
5	It greatly reduces contamination of the product so the sodium hydroxide produced will virtually be pure. If anions could pass through, the product would be contaminated with NaCl.
6	The aim is to extract pure caustic soda from the cathode compartment, hence it would not be appropriate to put brine here. It is necessary to add water to this compartment to maintain the amount of solution present. Brine is needed in the anode compartment to supply the Cl^- for the oxidation reaction.
7	a By evaporation b It would increase the costs since heat (and hence possibly electricity or a fuel) is required in the evaporation process to make it occur quicker.
8	For example, as fuel in a hydrogen-oxygen fuel cell and as a raw material for the production of ammonia.

Worksheet 16.1: Solutions Carbon compounds in the home	
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No.	Answer
1	$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_2\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$
2	Hydrocarbons such as butane are popular in many of these.
3	‘Essence’ usually stands for ethanol. Many cough syrups and food additives dissolve better in ethanol than water. The scrutiny occurs because of the possible misuse of these products as alcohol sources, and the effect of large quantities of ethanol-based products on driver blood alcohol levels.
4	$\text{C}_3\text{H}_4\text{O}_3$
5	$\text{CH}_2(\text{COOH})\text{C}(\text{OH})(\text{COOH})\text{CH}_2(\text{COOH})$
6	Acetone: CH_3COCH_3
7	Acetic acid: CH_3COOH
8	Lauryl sulfate is a commonly used surfactant (detergent molecule).
9	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$ ‘Polyunsaturated’ means that the hydrocarbon chain in the molecule contains two or more carbon–carbon double bonds.
10	$\text{CH}_2=\text{CHCH}=\text{CH}_2$

Worksheet 16.2: Solutions Modelling and naming carbon compounds	
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No.	Answer
1	The angle is just over 109° . The arrangement of the four bonds around the carbon atom is called a tetrahedral arrangement. See Figure 16.2a, page 374 in the textbook.
2	The H–C–H bond is close to 120° because there are 3 pairs of electrons repelling each other. If your molecule is correct, it will sit flat on the bench. ‘Planar’ means the molecule is flat or two-dimensional.
3	The arrangement of bonds around each carbon is tetrahedral. Each H–C–H bond has an angle just over 109° .
4	a C_4H_{10} c 7 b 3 d $n - 1$
5	Given there are two bonds, you might expect the length to be twice the length of one bond.
6	The zigzag nature of the carbon chain means the distance will be less than twice the length of a single bond. It will be approximately 1.63 times the length of your single bond.
7	Length = $0.814 (n - 1)$, where n = number of carbon atoms.
8	See Figure 16.14, page 388 in the textbook
9	Pentane has three isomers: pentane, 2-methylbutane and 2,2-dimethylpropane.
10	Hexane has five isomers: hexane, 2-methylpentane, 3-methylpentane, 2,3-dimethylbutane and 2,2-dimethylbutane. (See Figure 16.15, page 388 in the textbook.)
11	The sequence has been 1, 2, 3, 5 isomers for 3, 4, 5, 6 carbon atoms respectively, so perhaps heptane has 8?
12	a 2,3-dimethylbutane b 2-methylpropene c 3-methylpent-2-ene d dimethylpropane

Worksheet 16.3: Solutions Production of ethene	
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No.	Answer
1	Ethene is a planar molecule, with only 3 bonds around each carbon. It has no polarity, so the only intermolecular bonds are weak. This non-polar nature makes it insoluble in water, and the very weak intermolecular bonds result in a low boiling point.
2	The petroleum is heated to around 600°C in the absence of air. The components with boiling points higher than 600°C remain as a mixture of asphalts and tars. The vaporised components rise and gradually cool in a metal tower. At regular intervals in the tower there are horizontal trays. Components in the vapour with boiling temperatures such that they condense at the level of a tray, condense and are collected. This collected liquid (a fraction) contains a mixture of alkanes with similar boiling points. As the vapour rises up the tower, the trays collect fractions of decreasing boiling temperatures.
3	a Cracking reactions break large molecules into smaller ones. The degree of unsaturation is increased in the process. b Thermal and catalytic techniques are used.
4	a $C_4H_{10}(g) \rightarrow C_2H_4(g) + C_2H_6(g)$ b For example: $C_4H_{10}(g) \rightarrow C_3H_6(g) + CH_4(g)$
5	Increased pressure, increased temperature, use of a catalyst and increased concentration of reactants would all increase the rate.
6	The gases are heated to almost 1000°C (but the time spent in the cracking tubes is a fraction of a second). Low pressure (typically less than 1 atm) is used. (The problem of working with gases at pressures below atmospheric is overcome by the use of steam as an inert dilutant.)
7	$C_2H_6(g) \rightarrow C_2H_2(g) + 2H_2(g)$
8	Two molecules react to form one product. For example, the chlorine molecule ‘adds’ across the double bond: $CH_2CH_2(g) + Cl_2(g) \rightarrow CH_2ClCH_2Cl(g)$
9	a Polyethene $(CH_2)_n$ b CH_3CH_2Cl c CH_3CH_3 d CH_3CH_2OH e CH_2BrCH_2Br
10	petroleum $\xrightarrow{\text{fractional distillation}}$ heavier fractions $\xrightarrow{\text{cracking process}}$ ethene + branched chain alkanes